

3D Imaging of the Pion on a Fine Lattice

Based on arXiv:2504.04625

Co-authors: Dennis Bollweg, Xiang Gao, Swagato Mukherjee, and Yong Zhao

Jinchen He

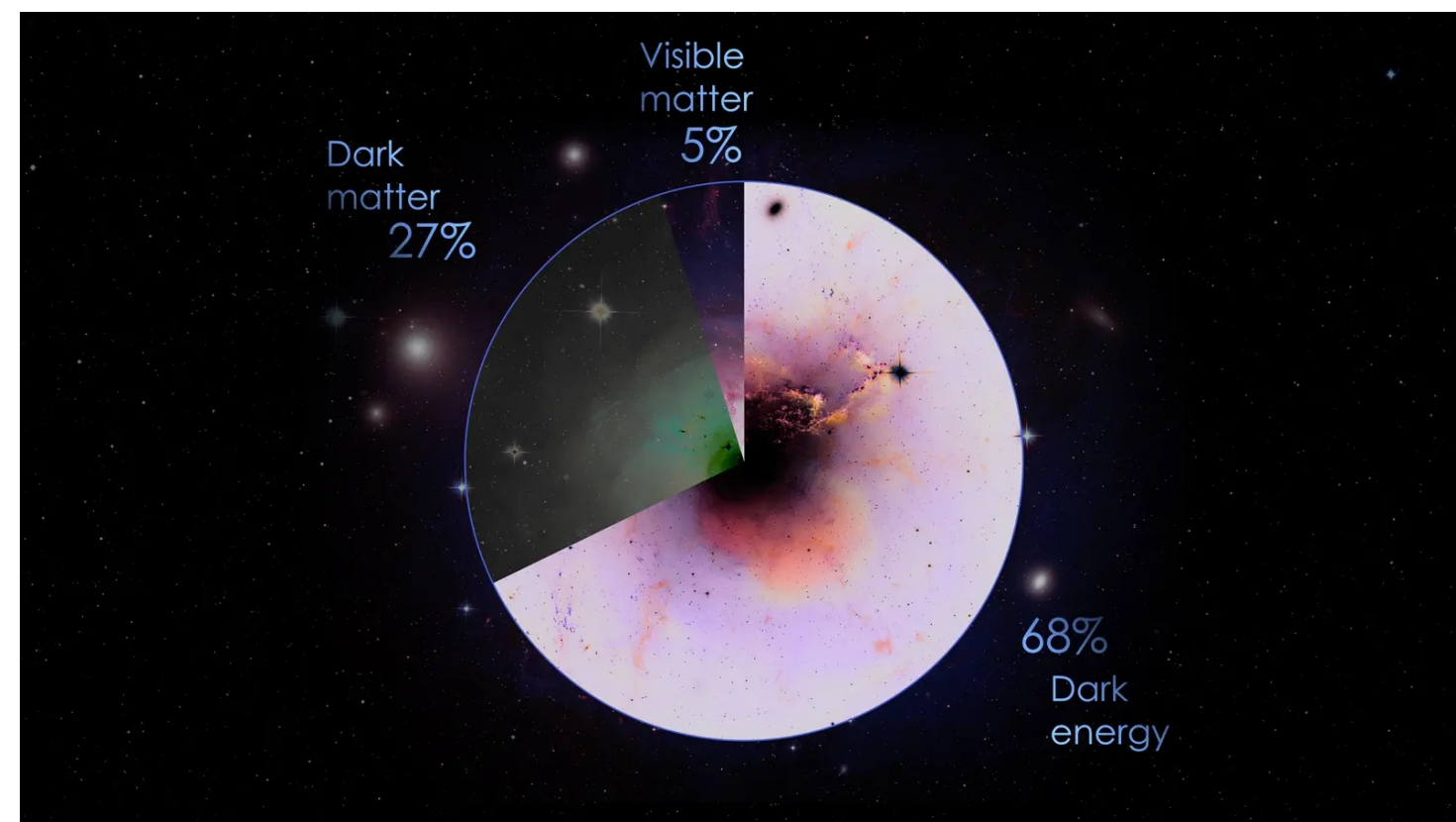
New Perspectives 2025

2025/07

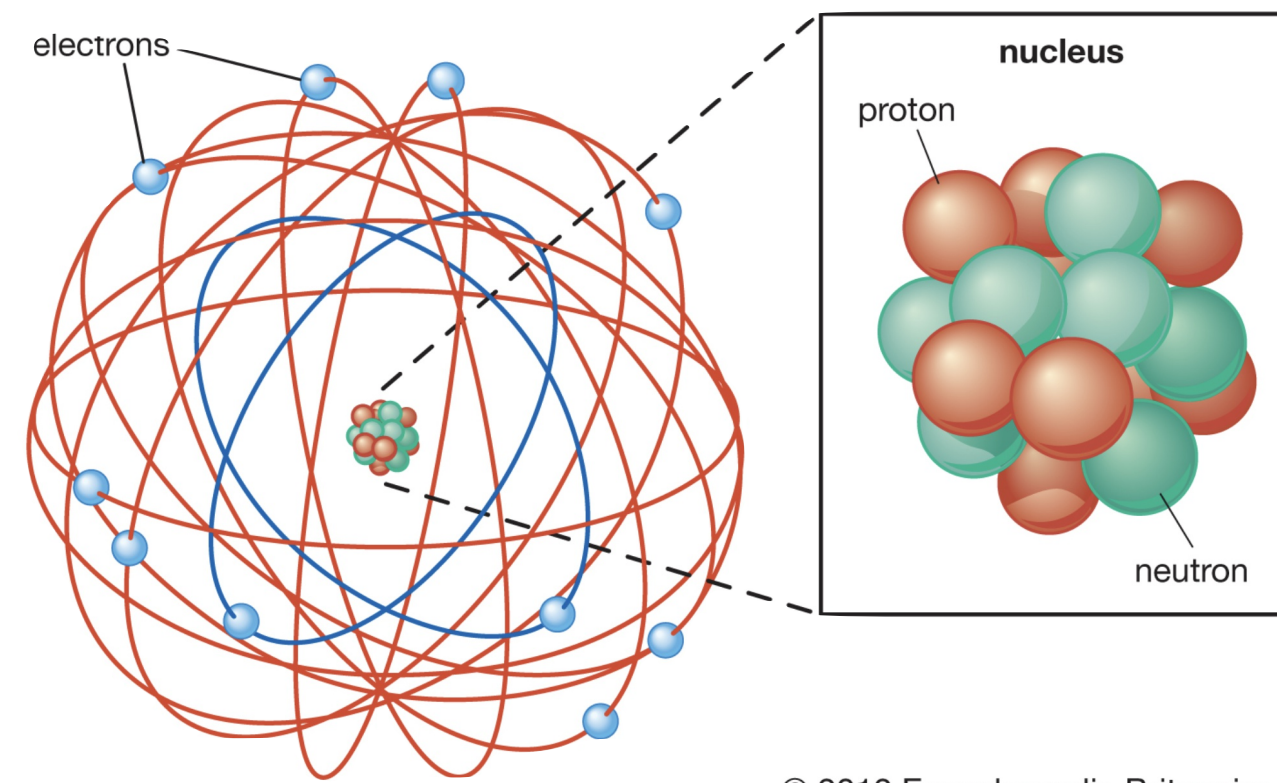


Visible Universe

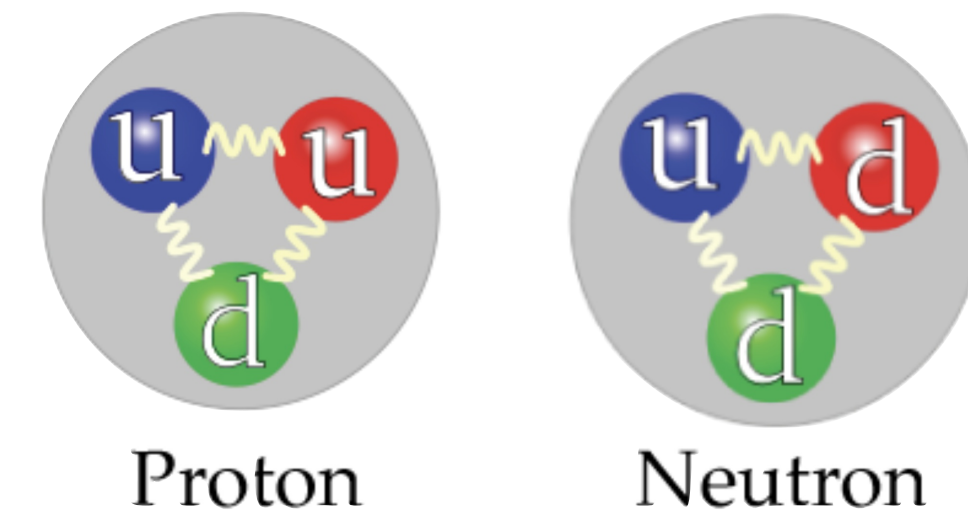
- Only 5% of the universe is visible. *Spergel, David N. "The dark side of cosmology: Dark matter and dark energy." Science 347.6226 (2015): 1100-1102.*
- The visible universe is made up of protons and neutrons, the inner structure of hadrons are sophisticated if we step closer.



Cr. NASA's Goddard Space Flight Center



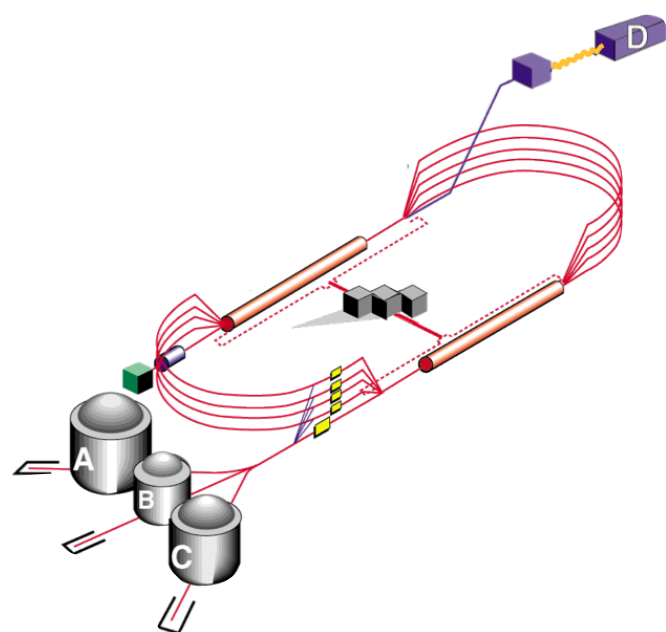
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- Many experiments have been designed to probe the internal structure of hadrons.

P.S. The list of experiments here is not complete.

CEBAF(JLab)



Cr. Dave Gaskell

HERA



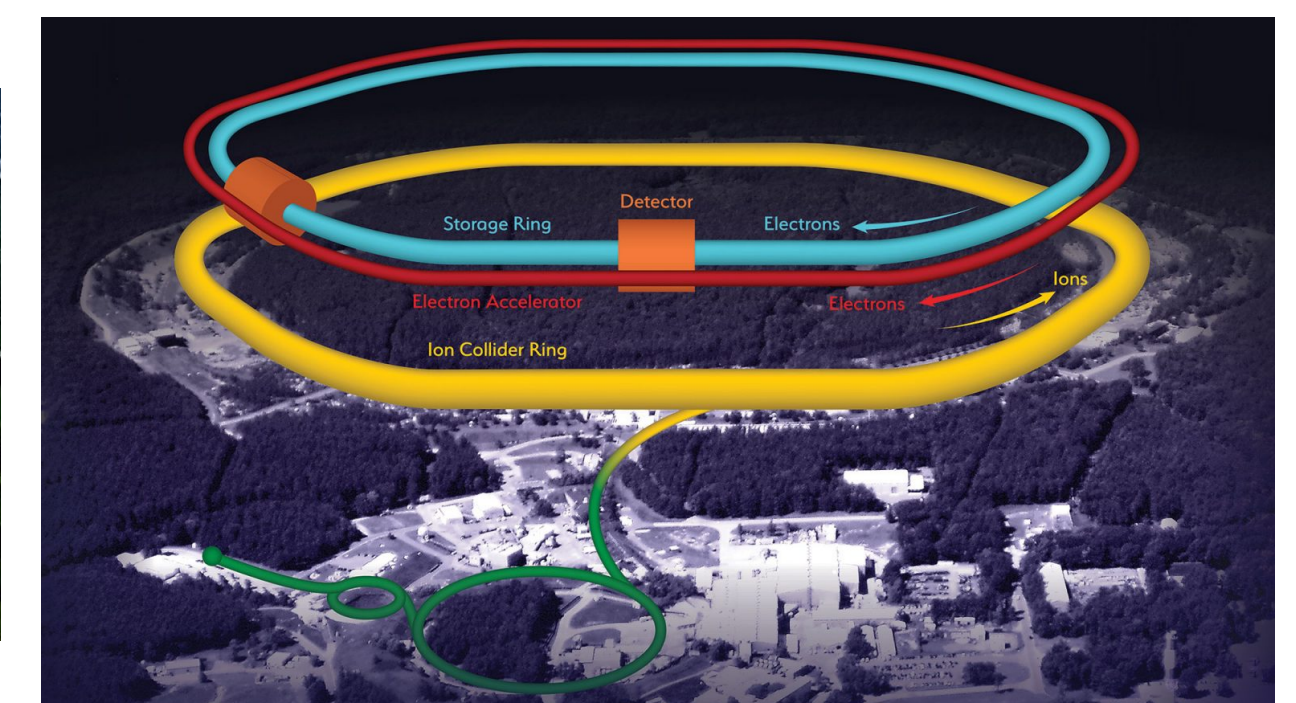
Cr. DESY

LHC



Cr. CERN

EIC



Cr. BNL

3D Imaging of Hadron

x is the momentum fraction in the longitudinal (hadron momentum) direction.

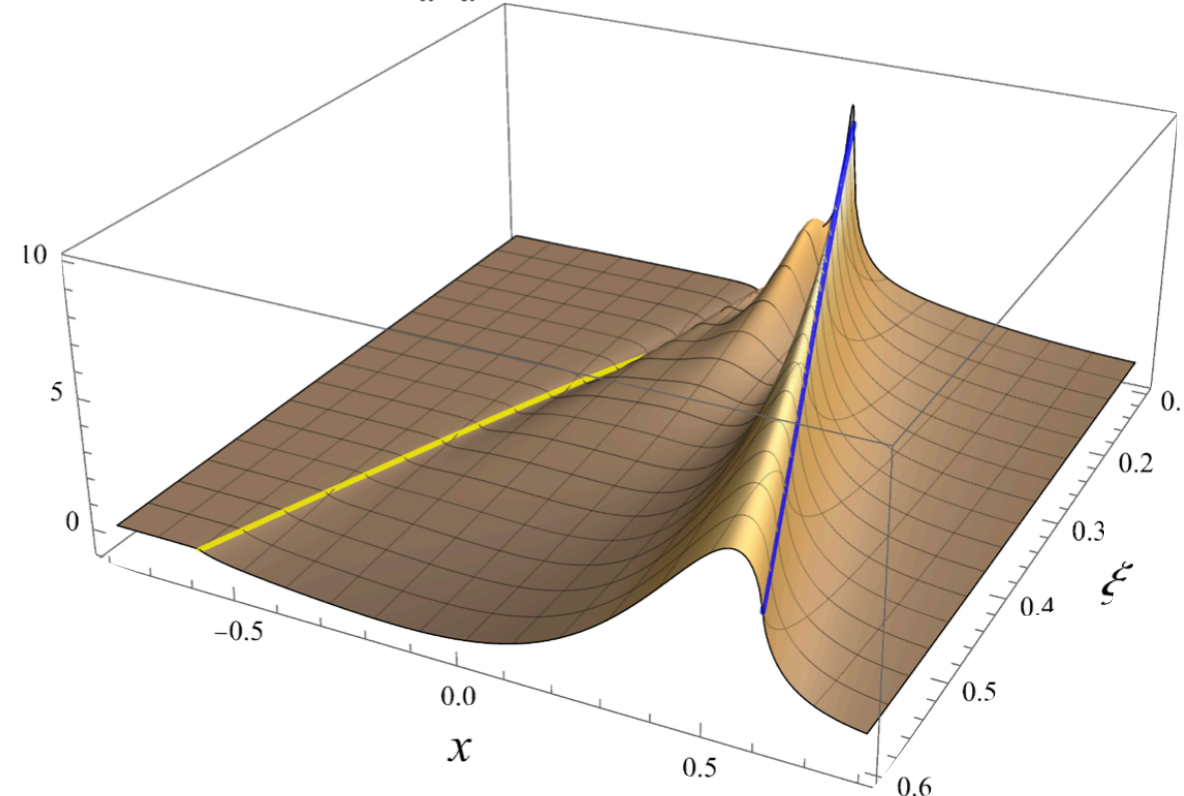
Wigner Distribution / GTMD

$$W(x, \vec{r}_\perp, \vec{k}_\perp)$$

$$\int d^2\vec{k}_\perp$$

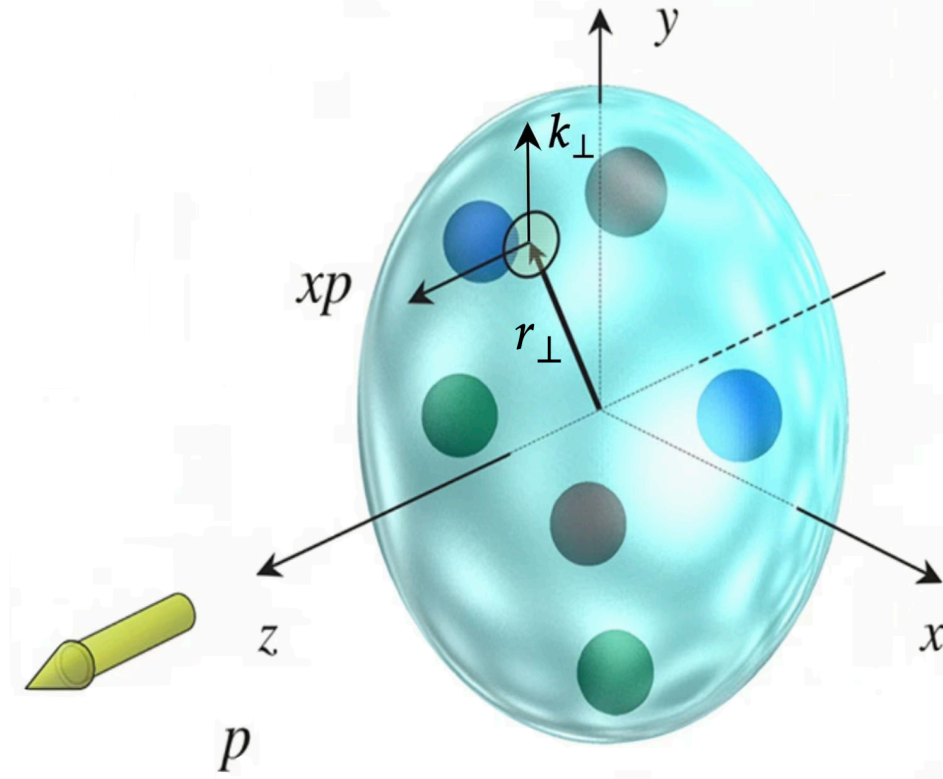
Generalized Parton Distributions (GPDs)

The isovector GPD H_{u-d} at $-t = 0.69 \text{ GeV}^2$ tuned with DA terms



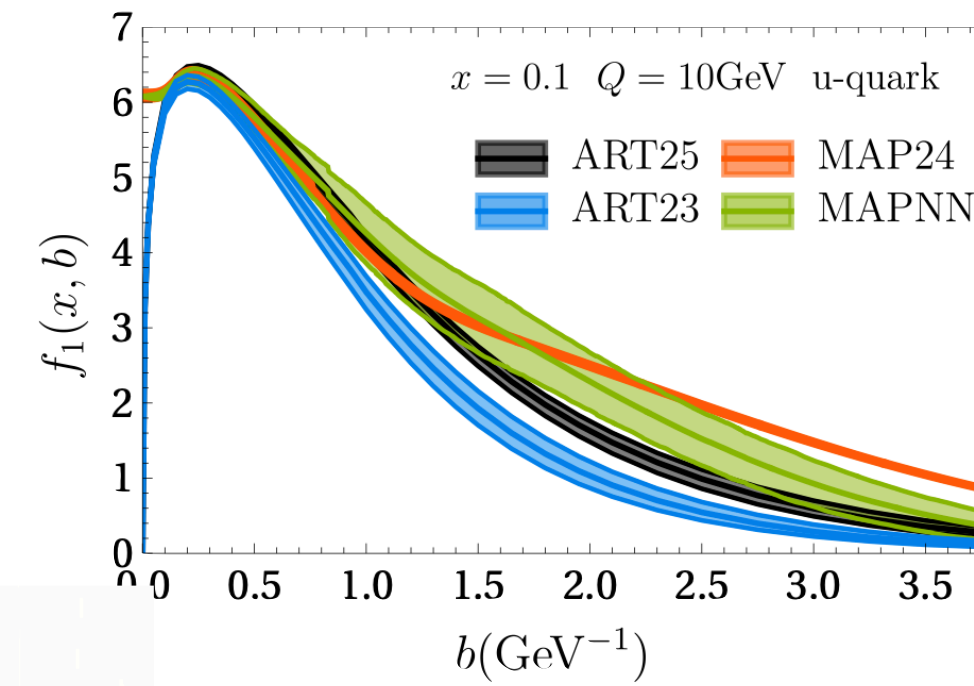
Y. Guo, et al., JHEP 09 (2022)

$$\int d^2\vec{r}_\perp$$

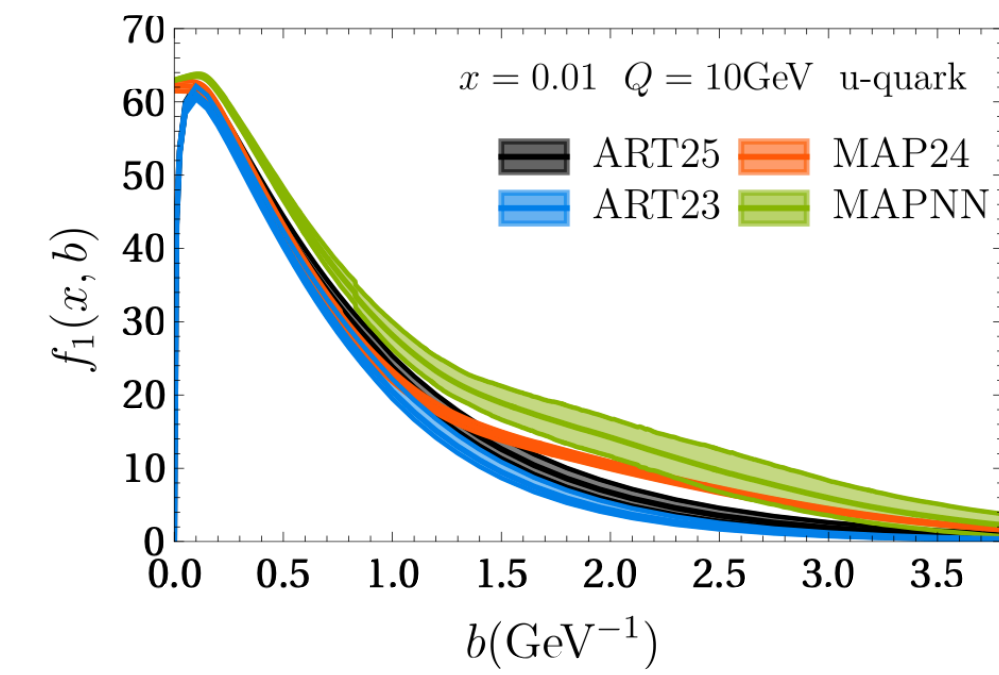


$$\int d^2\vec{r}_\perp$$

Transverse-Momentum-Dependent distributions (TMDs)



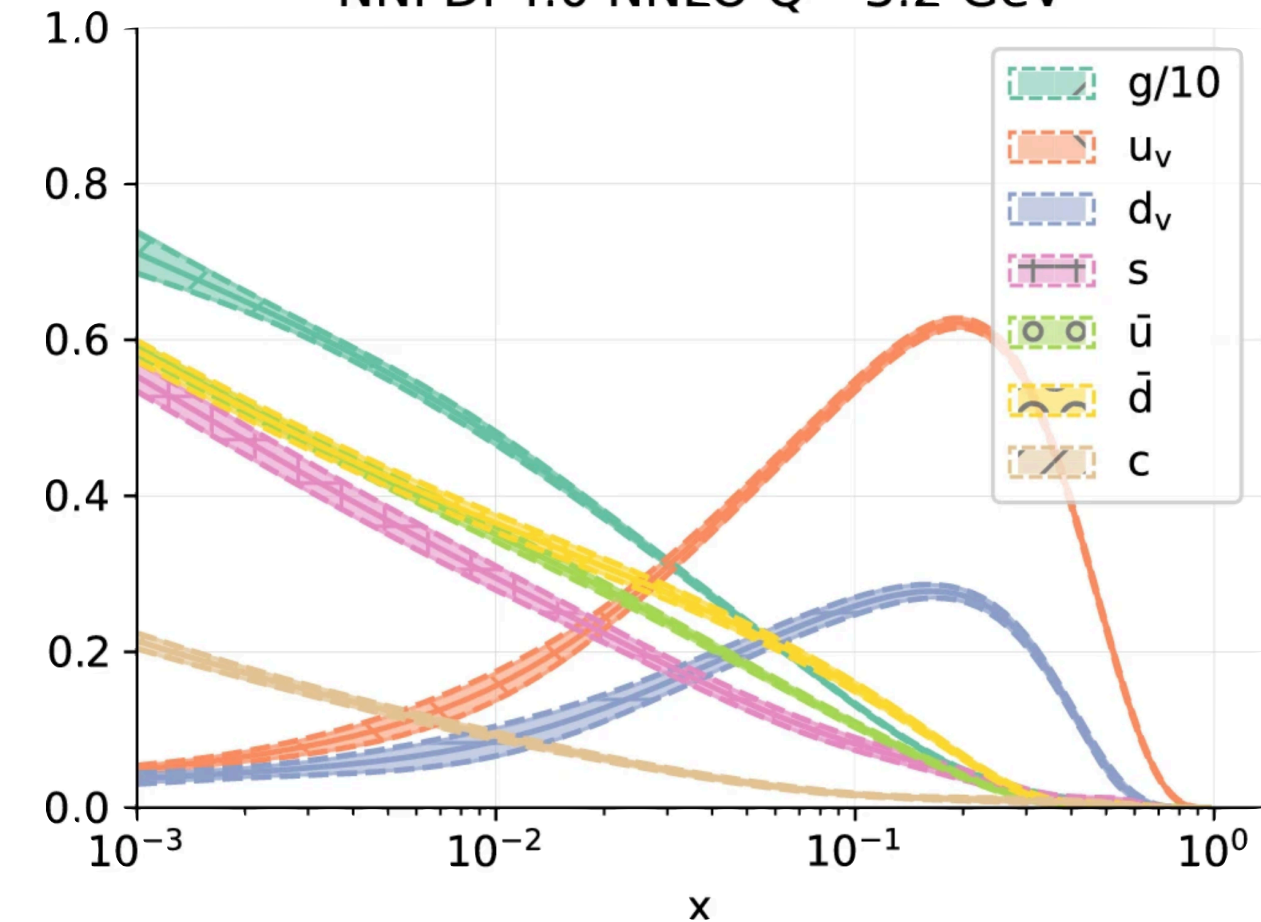
V. Moos, et al., 2503.11201 (2025)



$$\int d^2\vec{k}_\perp$$

Parton Distribution Functions (PDFs)

NNPDF4.0 NNLO $Q = 3.2 \text{ GeV}$

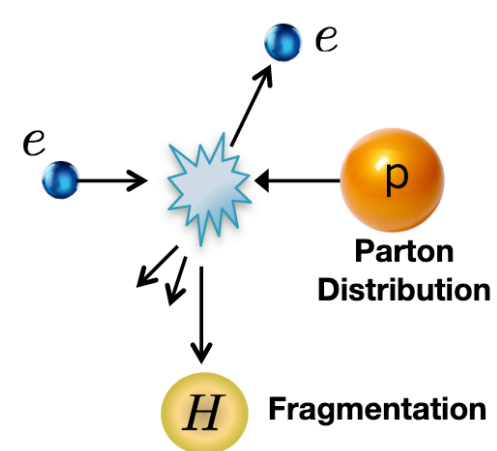


R. D. Ball, et al. [NNPDF], Eur. Phys. J. C 82 (2022)

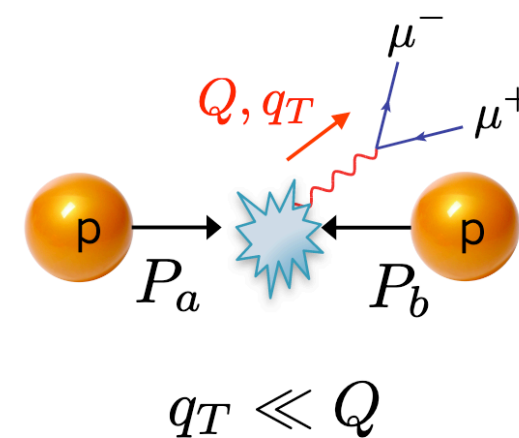
TMDs in Experiments

- **TMDPDFs**: the distribution densities of finding a parton carrying a longitudinal momentum fraction x and transverse momentum $k_\perp$ in a hadron;
- TMD processes are important processes in high energy collisions, like Drell-Yan process on LHC and Semi-Inclusive DIS on EIC;

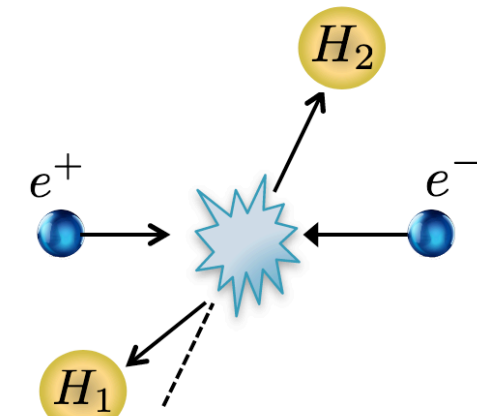
Semi-Inclusive DIS



Drell-Yan



Dihadron in e+e-

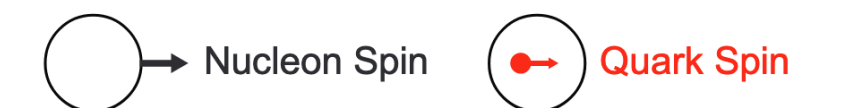


$$\sigma_{\text{DY}} \propto \left| \sum_{i,j} \bar{u}_i \gamma_\mu u_j \right|^2 \approx \left| \sum_{i,j} \bar{u}_i \gamma_\mu u_j \right|^2 \otimes \left| \sum_{i,j} \bar{u}_i \gamma_\mu u_j \right|^2 \otimes \left| \sum_{i,j} \bar{u}_i \gamma_\mu u_j \right|^2$$

$$\frac{d\sigma_{H_a+H_b \rightarrow l\bar{l}+X}}{dQ^2 dY d^2\vec{q}_T} = \frac{4\pi\alpha^2}{3N_c Q^2 S} \sum_i e_i^2 \int d^2\vec{k}_{a\perp} d^2\vec{k}_{b\perp} \delta^{(2)}(\vec{q}_T - \vec{k}_{a\perp} - \vec{k}_{b\perp})$$

$$\times f_1(i/H_a)(x_a, \vec{k}_{a\perp}) \times f_1(\bar{l}/H_b)(x_b, \vec{k}_{b\perp})$$

Leading Quark TMDPDFs



		Quark Polarization		
		Un-Polarized (U)	Longitudinally Polarized (L)	Transversely Polarized (T)
Nucleon Polarization	U	$f_1 = \text{Unpolarized}$		$h_1^\perp = \text{Boer-Mulders}$
	L		$g_1 = \text{Helicity}$	$h_{1L}^\perp = \text{Worm-gear}$
	T	$f_{1T}^\perp = \text{Sivers}$	$g_{1T}^\perp = \text{Worm-gear}$	$h_1 = \text{Transversity}$ $h_{1T}^\perp = \text{Pretzelosity}$

R. Boussarie, et al., 2304.03302 (2023)

Lattice QCD

- Path integral formalism

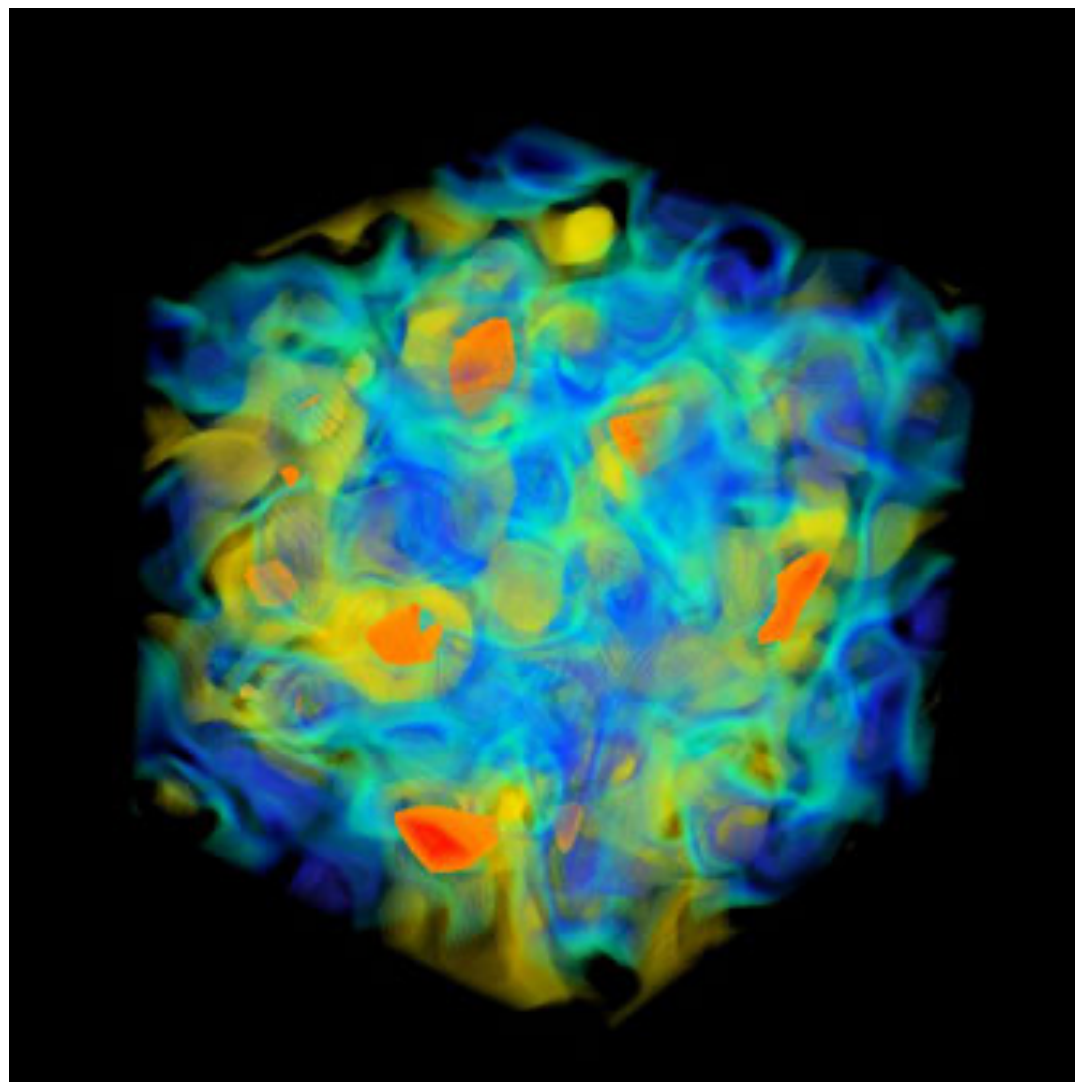
$$Z = \int \mathcal{D}A \mathcal{D}\psi \mathcal{D}\bar{\psi} e^{iS_{\text{QCD}}[A,\psi,\bar{\psi}]} \xrightarrow[t \rightarrow -it_E]{\substack{\text{Wilson link} \\ U_\mu(x) = e^{igaA^\mu(x)}}} Z_E = \int \mathcal{D}U e^{-S_E^g[U]} \int \mathcal{D}\psi \mathcal{D}\bar{\psi} e^{-\bar{\psi}M[U]\psi} = \int \mathcal{D}U e^{-S_E^g[U]} \det M[U]$$

Wick rotation

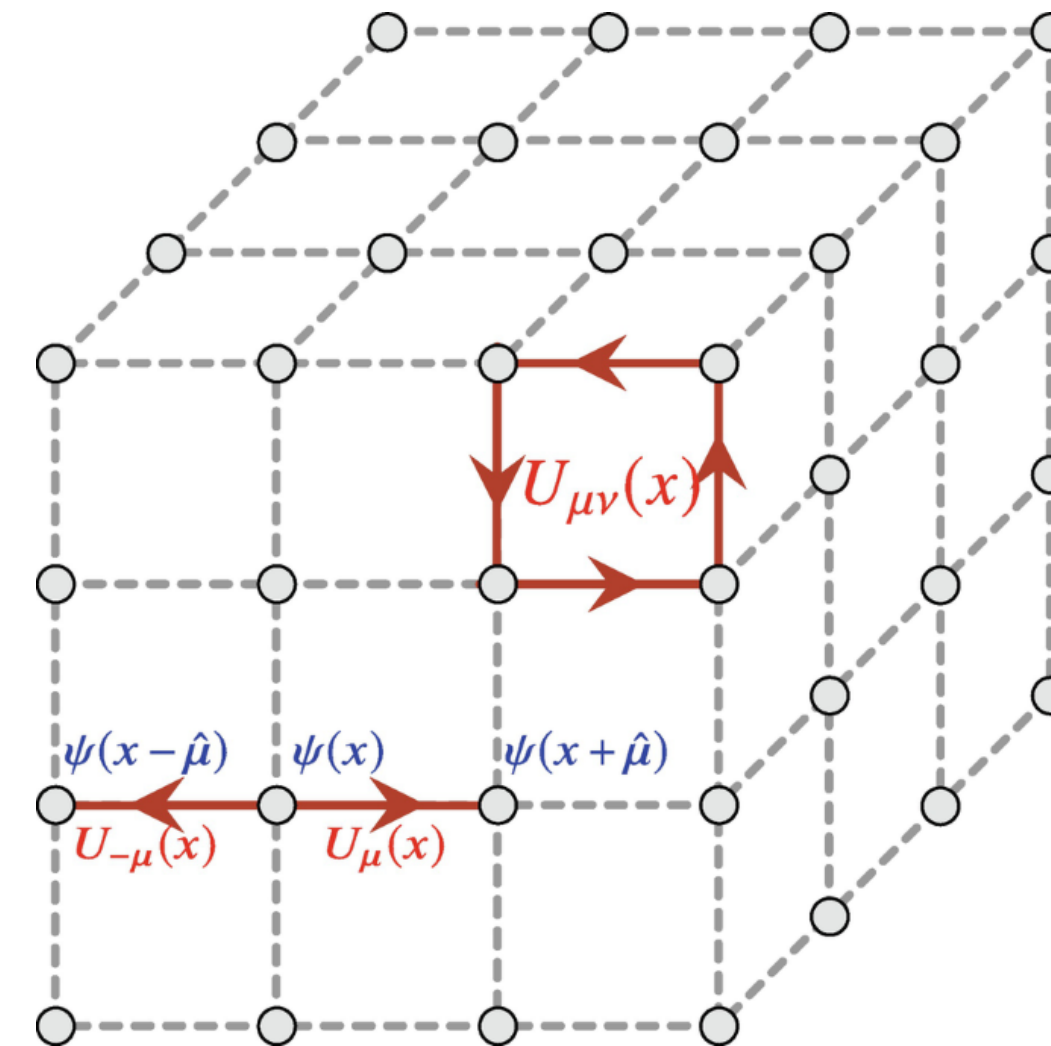
Sampling probability for configuration U

- Monte Carlo sampling

$$\langle \hat{O} \rangle = \frac{1}{Z_E} \int \mathcal{D}U \mathcal{D}\psi \mathcal{D}\bar{\psi} \hat{O} e^{-S_E^{\text{QCD}}[A,\psi,\bar{\psi}]} = \frac{1}{N} \sum_{i=1}^N O[U^{(i)}]$$



Cr. BNL



Cr. Olaf Kaczmarek

Large-Momentum Effective Theory (LaMET)

- TMDPDF is defined from a light-cone correlator in a hadron, which is Lorentz invariant.

Parton model

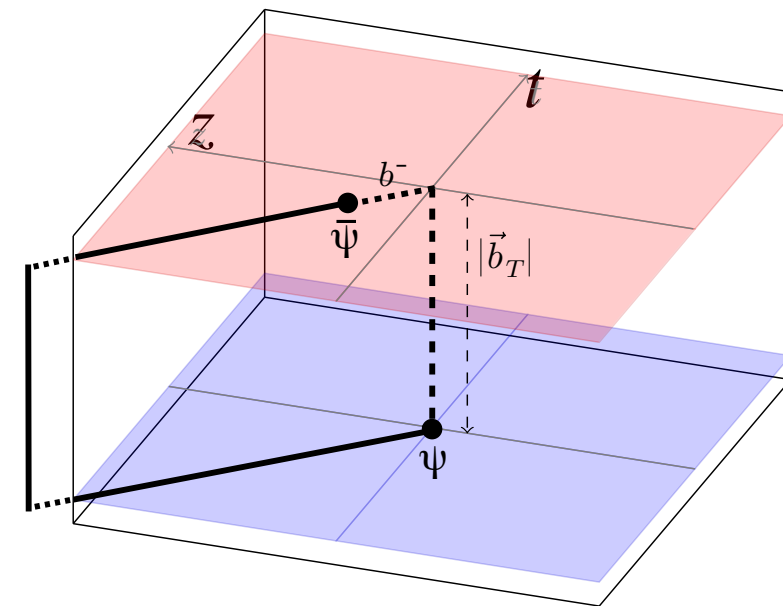
$$f(x, b_{\perp}, \dots) = \int_{-\infty}^{\infty} \frac{db^-}{2\pi} e^{-ib^-(xP^+)} \left\langle P \left| \bar{\psi}(b^{\mu}) W_{\square}(b^{\mu}, 0) \frac{\gamma^+}{2} \psi(0) \right| P \right\rangle \longleftrightarrow \left\langle |\vec{P}| = \infty \left| O(t=0) \right| |\vec{P}| = \infty \right\rangle$$

- Define a quasi distribution with large-momentum states and time-independent operators.

Different orders of limit, but pert.

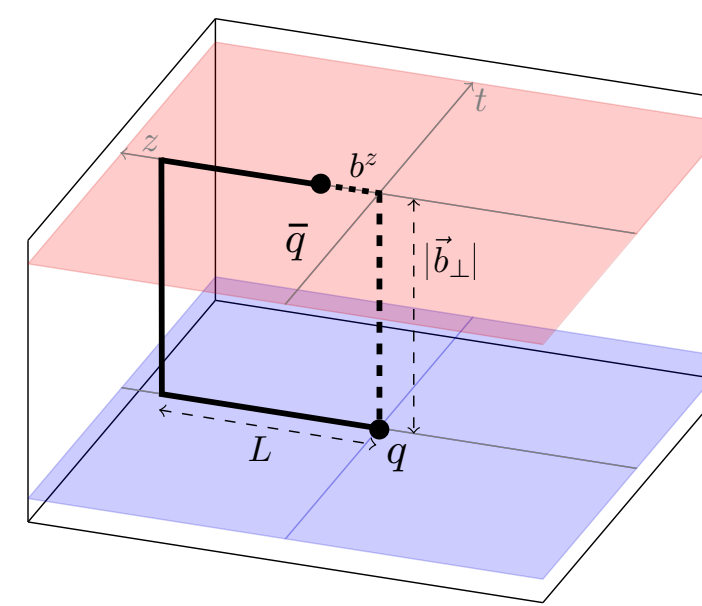
$$\tilde{f}_{\Gamma}^0(x, b_{\perp}, P^z, \mu) = P^z \int \frac{dz}{2\pi} e^{iz(xP^z)} \frac{1}{2P^t} \langle P | \bar{\psi}_0(z, b_{\perp}) W_{\square}(z, b_{\perp}; 0) \Gamma \psi_0(0) | P \rangle, \quad \Lambda_{\text{QCD}} \ll |\vec{P}| \ll \frac{\pi}{a}$$

Large-momentum expansion



X. Ji, *Phys.Rev.Lett.* 110 (2013)
X. Ji, et al., *Rev.Mod.Phys.* 93 (2021)
X. Ji, *Nucl. Phys. B* 1007 (2024)

Lorentz boost & Matching



Light-cone distribution:

Cannot be directly calculated on the lattice

Quasi distribution:

Directly calculable on the lattice

$H_f(x, P^z; \mu) = |C_{\text{TMD}}(xP^z; \mu)|^2$ is the TMD hard kernel for matching.

- LaMET enables us to obtain the precision-controlled x-distribution of TMDs in $x \in [x_{\min}, x_{\max}]$.

Collins-Soper scale: $\zeta \sim 2(xP^+)^2$

$$\sqrt{S_I(b_{\perp}; \mu)} \cdot \tilde{f}_{\Gamma}(x, b_{\perp}, P^z; \mu) = f(x, b_{\perp}; \mu, \zeta) H_f(x, P^z; \mu) \exp \left[\frac{1}{2} \ln \frac{(2xP^z)^2}{\zeta} \gamma^{\overline{\text{MS}}}(b_{\perp}; \mu) \right] + \text{Power corrections}$$

Coulomb Gauge Method

- Define a quasi distribution in CG without Wilson line: *X. Gao, W. Y. Liu and Y. Zhao, PRD 109 (2024)*

Y. Zhao, PRL 133 (2024)

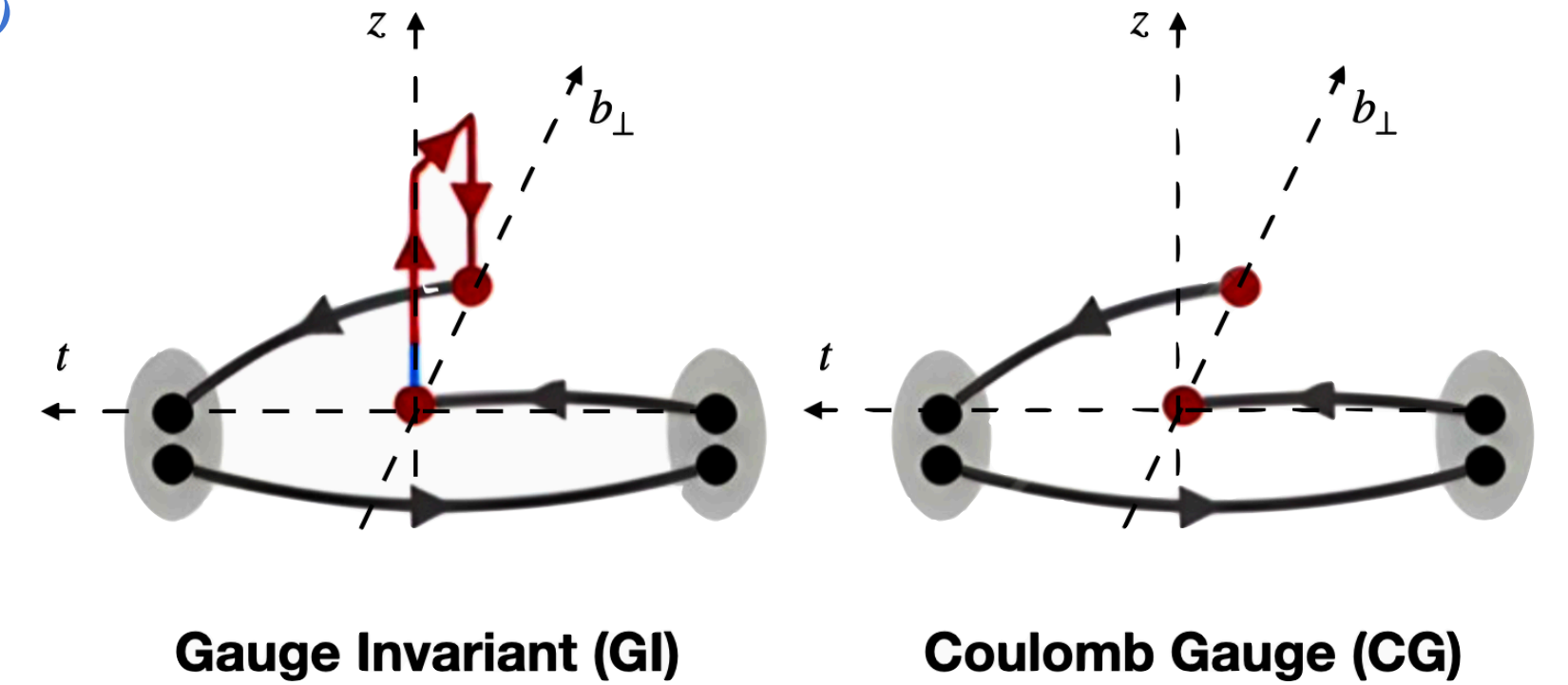
$$\tilde{f}_{\text{CG}}^0(x, b_{\perp}, P^z, \mu) = P^z \int \frac{dz}{2\pi} e^{iz(xP^z)} \frac{1}{2P^t} \langle P | \bar{\psi}_0(z, b_{\perp}) \Gamma \psi_0(0) | P \rangle \Big|_{\vec{\nabla} \cdot \vec{A} = 0}$$

- Why choose CG?

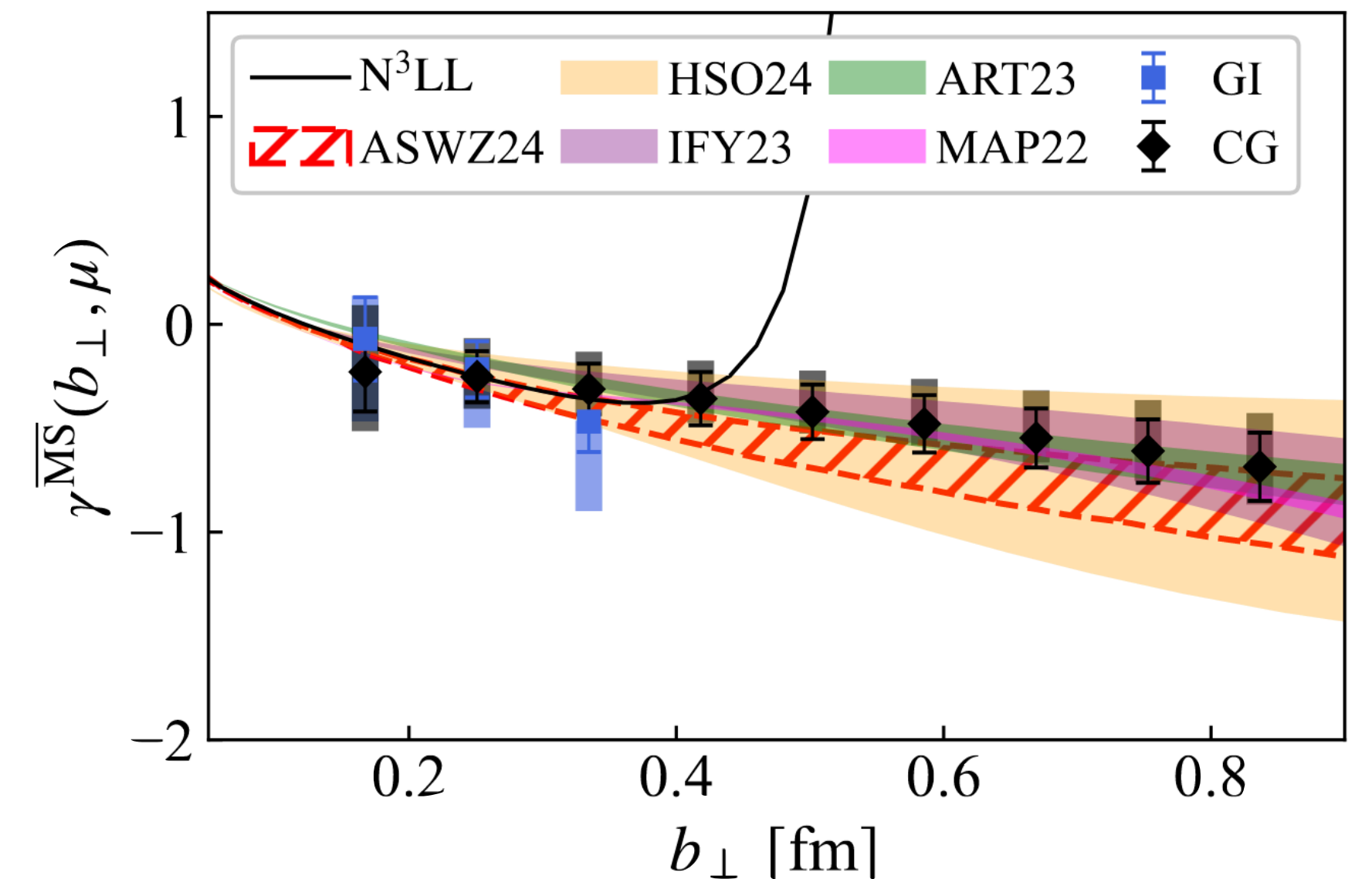
X. Ji, Y. S. Liu, Y. Liu, J. H. Zhang and Y. Zhao, RMP 93 (2021)

- $\vec{\nabla} \cdot \vec{A} = 0$ becomes $A^+ = 0$ in the infinite boost, so the quasi distribution in CG belongs to the universality class in LaMET;
- No linear divergence from the Wilson link, improving the SNR significantly;
- Simplified renormalization $\bar{\psi}_0(z, b_{\perp}) \Gamma \psi_0(0) = Z_{\psi}(a) [\bar{\psi}(z, b_{\perp}) \Gamma \psi(0)]$;
- Larger off-axis momenta (3D rotational symmetry).

The results in CG and GI are consistent with the same lattice setup;
Compared with the GI method, CG method has much better SNR.



Collins-Soper kernel in CG v.s. GI



D. Bollweg, et al., Phys.Lett.B 852 (2024)

Lattice Setup

- 2+1 flavor HISQ ensemble by HotQCD with volume $L_s \times L_t = 48^3 \times 64$;

- Lattice spacing is $a = 0.06$ fm;

- Pion mass of sea quark: $m_\pi^{\text{sea}} = 160$ MeV;

- Pion mass of valence quark for quasi-TMD: $m_\pi^{\text{val}} = 300$ MeV;

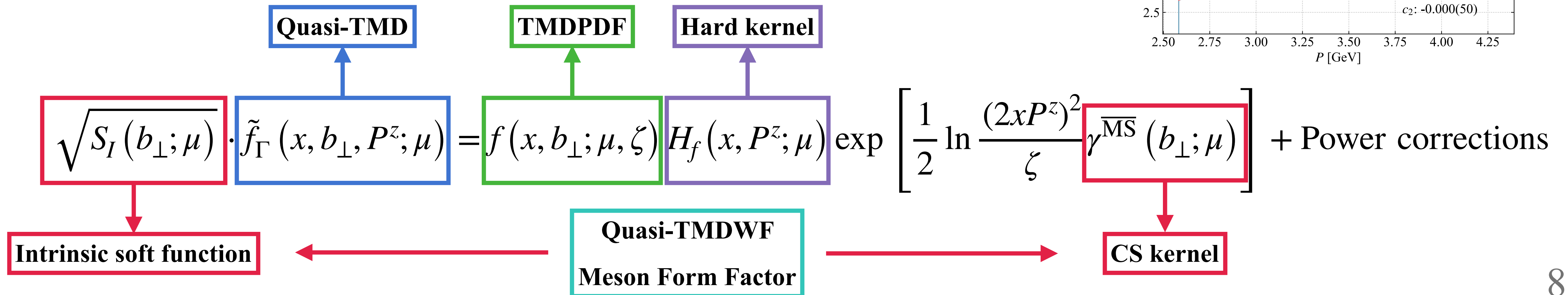
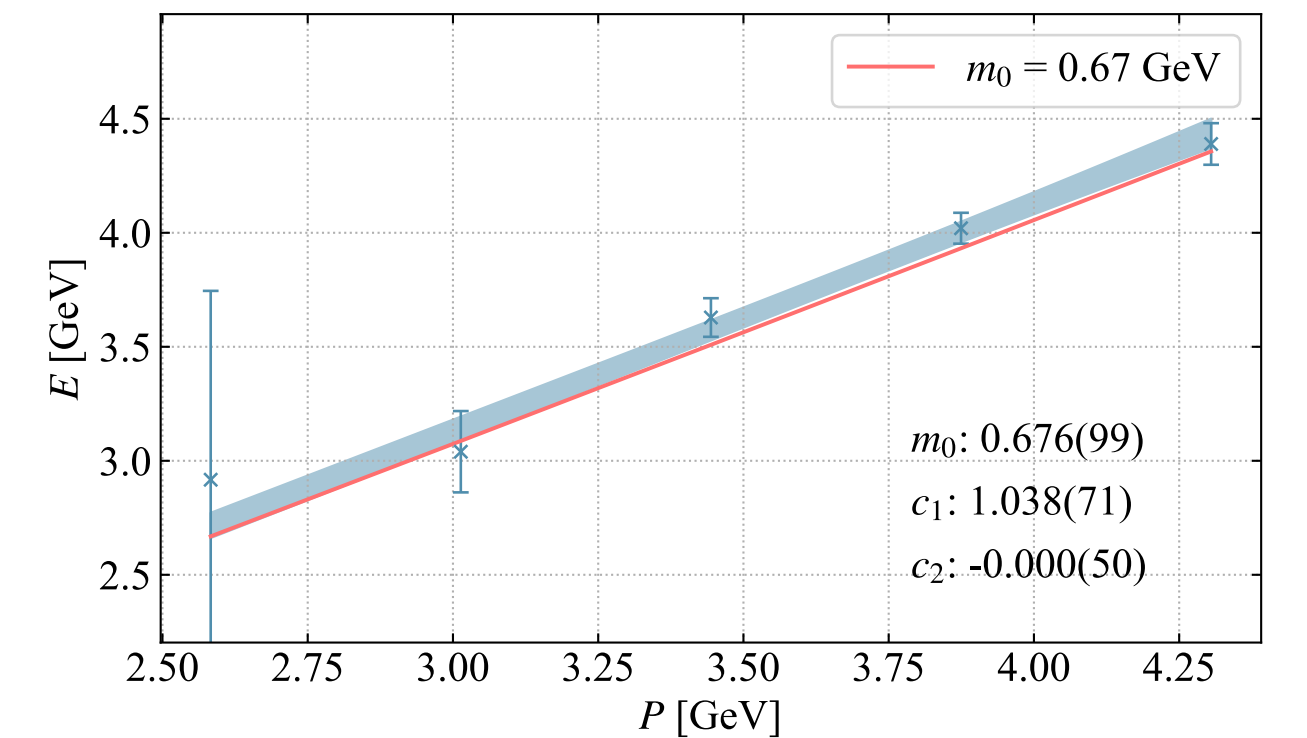
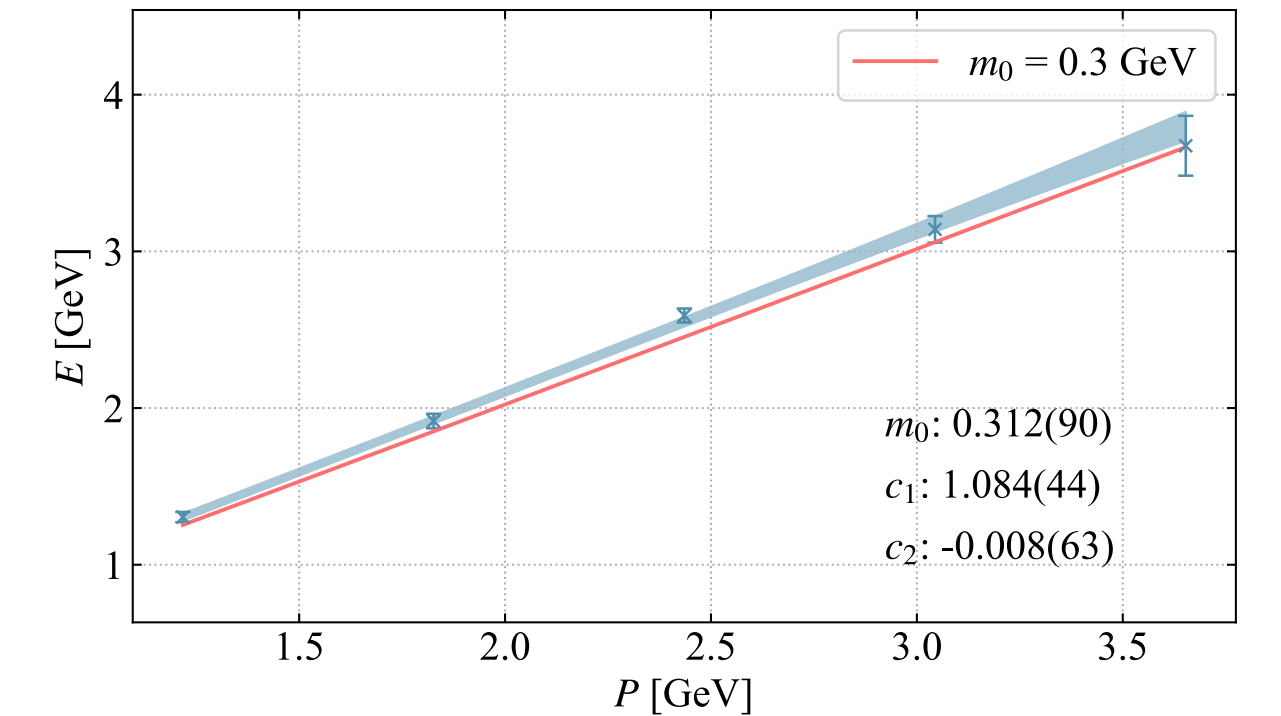
- Off-axis ($\vec{n} = (1,1,0)$) hadron momenta for quasi-TMD: 1.83 GeV, 2.43 GeV and 3.04 GeV;

- Pion mass of valence quark for qTMDWF and meson form factor: $m_\pi^{\text{val}} = 670$ MeV;

- On-axis hadron momenta for qTMDWF: 3.44 GeV, 3.87 GeV and 4.30 GeV;

- On-axis hadron momenta for meson form factor: 2.58 GeV.

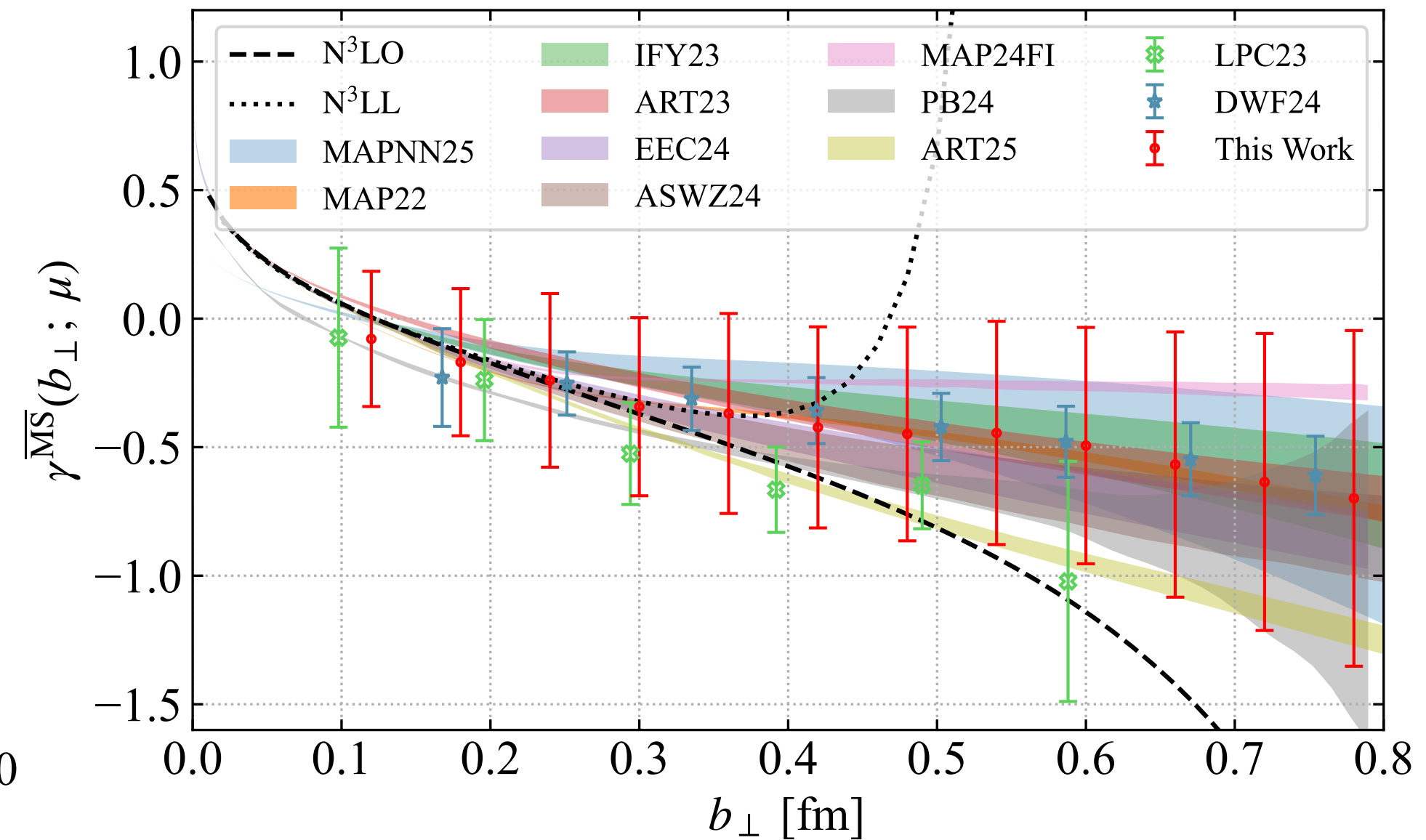
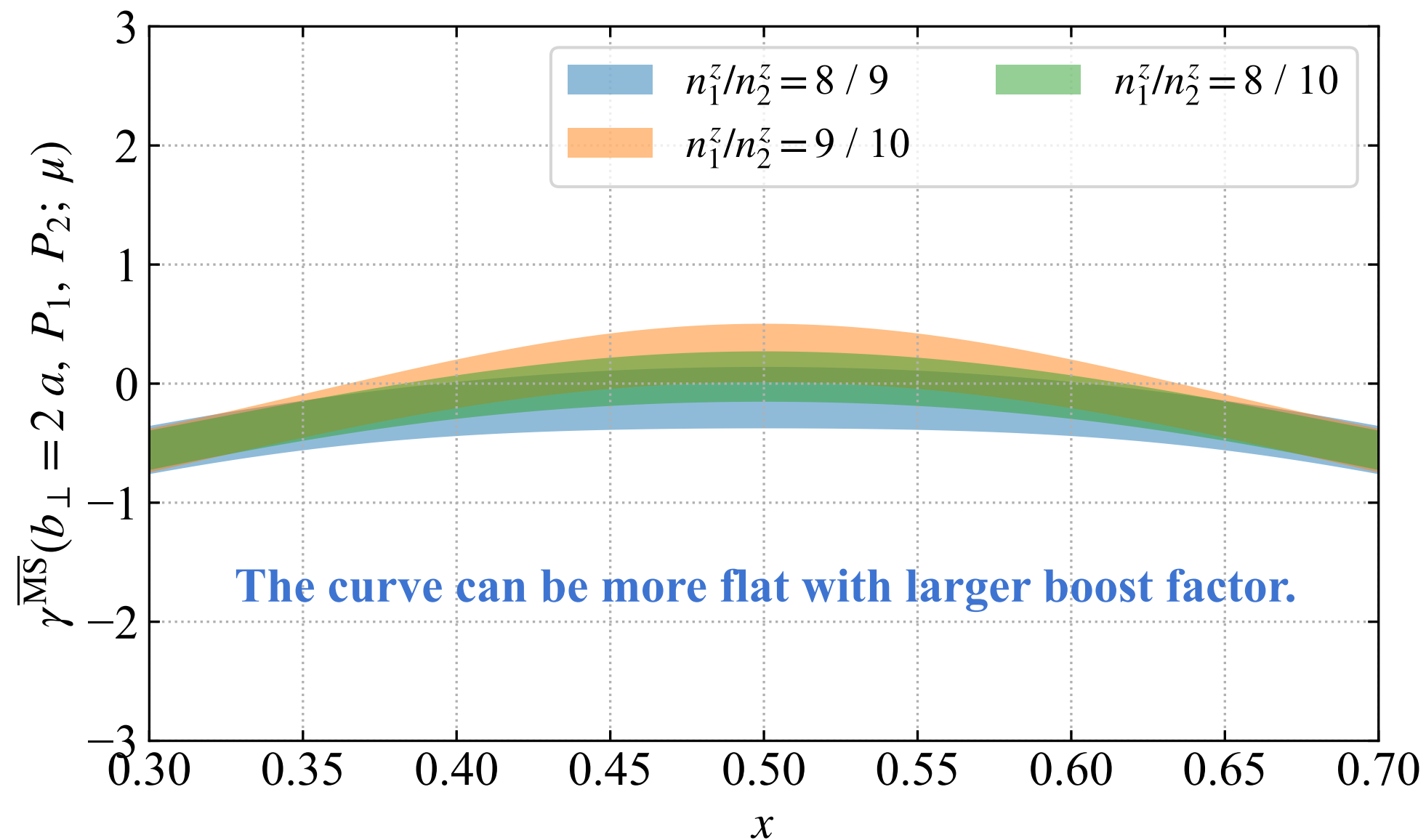
Dispersion relation: $E^2 = m_0^2 + c_1 P^2 + c_2 a^2 P^4$



Collins-Soper Kernel

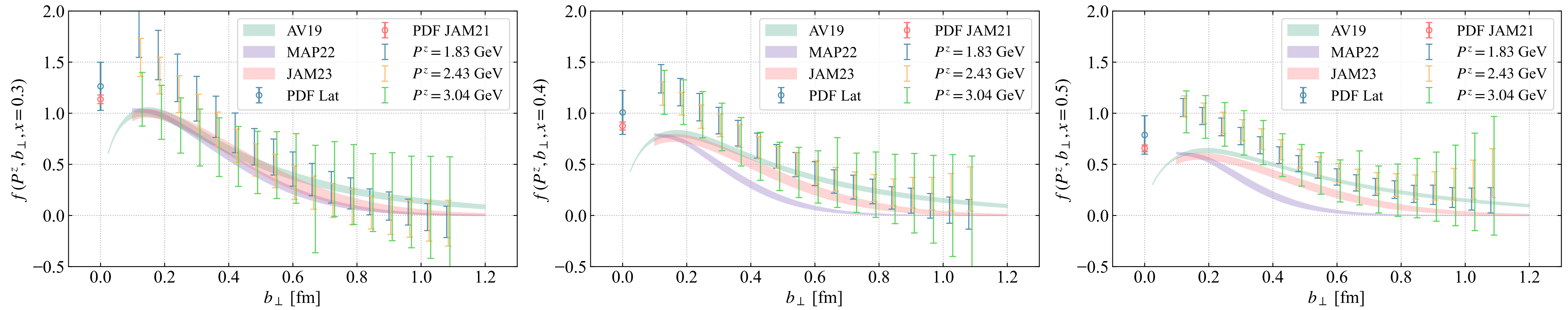
$$\gamma^{\overline{\text{MS}}}(b_{\perp}, P_1, P_2; \mu) = \frac{1}{\ln(P_2/P_1)} \ln \frac{H_{\phi}(x, \bar{x}, P_1; \mu) \tilde{\phi}_{\gamma^z \gamma^5}(x, b_{\perp}, P_2; \mu)}{H_{\phi}(x, \bar{x}, P_2; \mu) \tilde{\phi}_{\gamma^z \gamma^5}(x, b_{\perp}, P_1; \mu)}$$

$H_{\phi}(x, \bar{x}, P^z; \mu) = C_{\text{TMD}}(xP^z; \mu) \cdot C_{\text{TMD}}(\bar{x}P^z; \mu)$ is the TMD hard kernel for matching.



- DWF24 is another lattice calculation using the CG method on chirally symmetric domain-wall fermion configurations;
- There is a notable tension among recent results in phenomenology (MAP24FI & ART25);
- Both remain consistent with this work due to the large uncertainty;
- The large uncertainty is mainly caused by the **small Lorentz boost factor** at such a heavy pion mass ($m_{\pi}^{\text{val}} = 670$ MeV);

Pion TMDPDF in the $b_\perp$ Space



A. Vladimirov, JHEP 10 (2019); M. Cerutti, et al. (MAP), Phys.Rev.D 107 (2023); P. C. Barry, et al. (JAM) Phys.Rev.D 108 (2023)

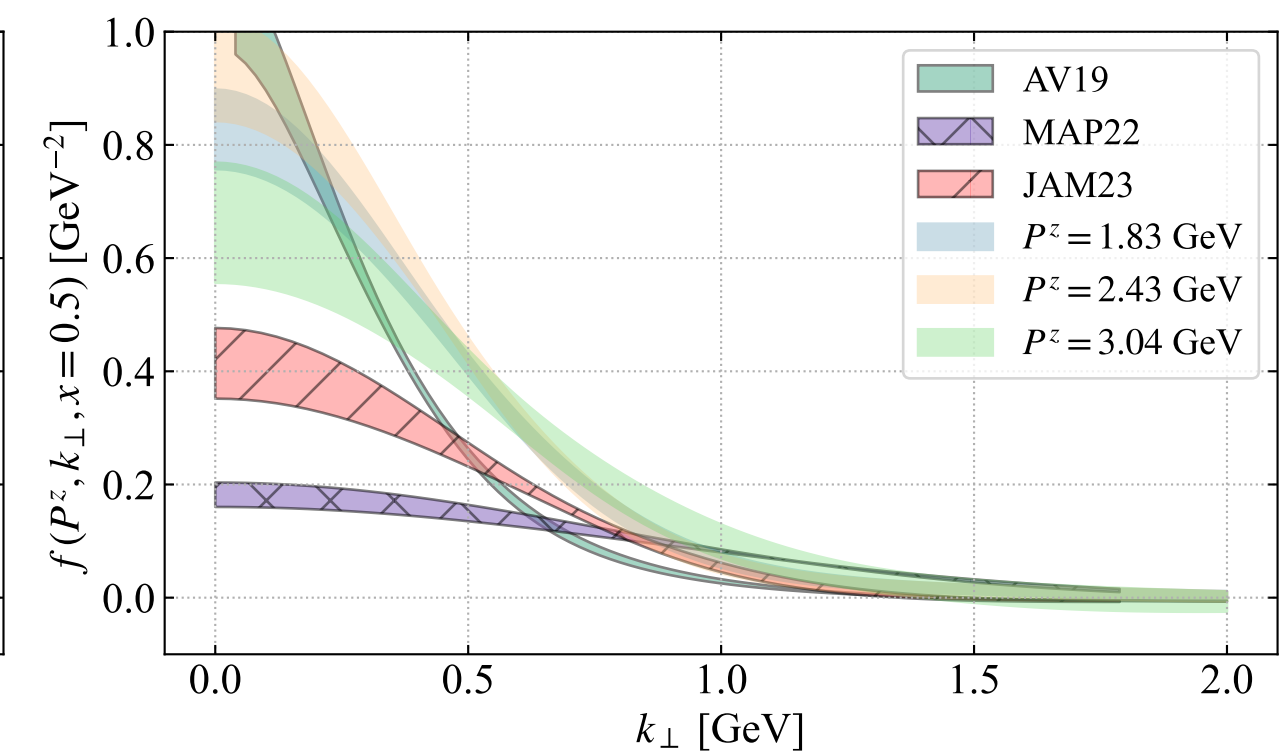
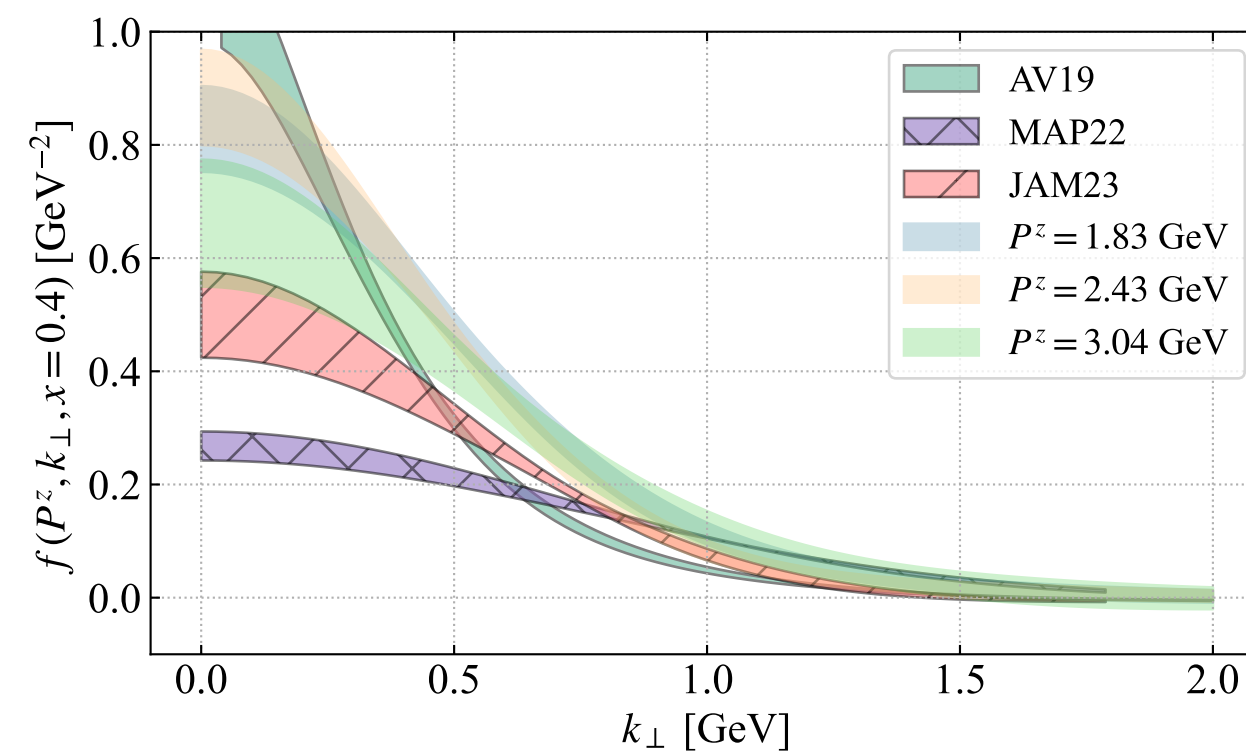
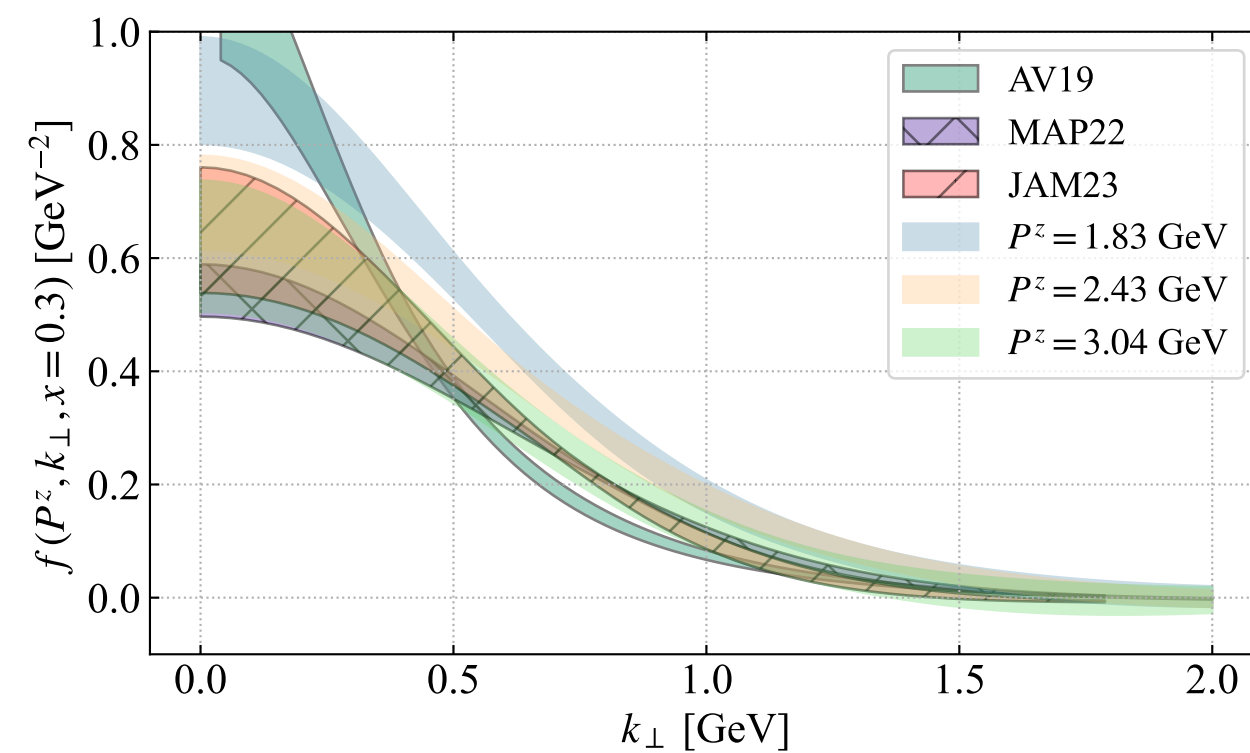
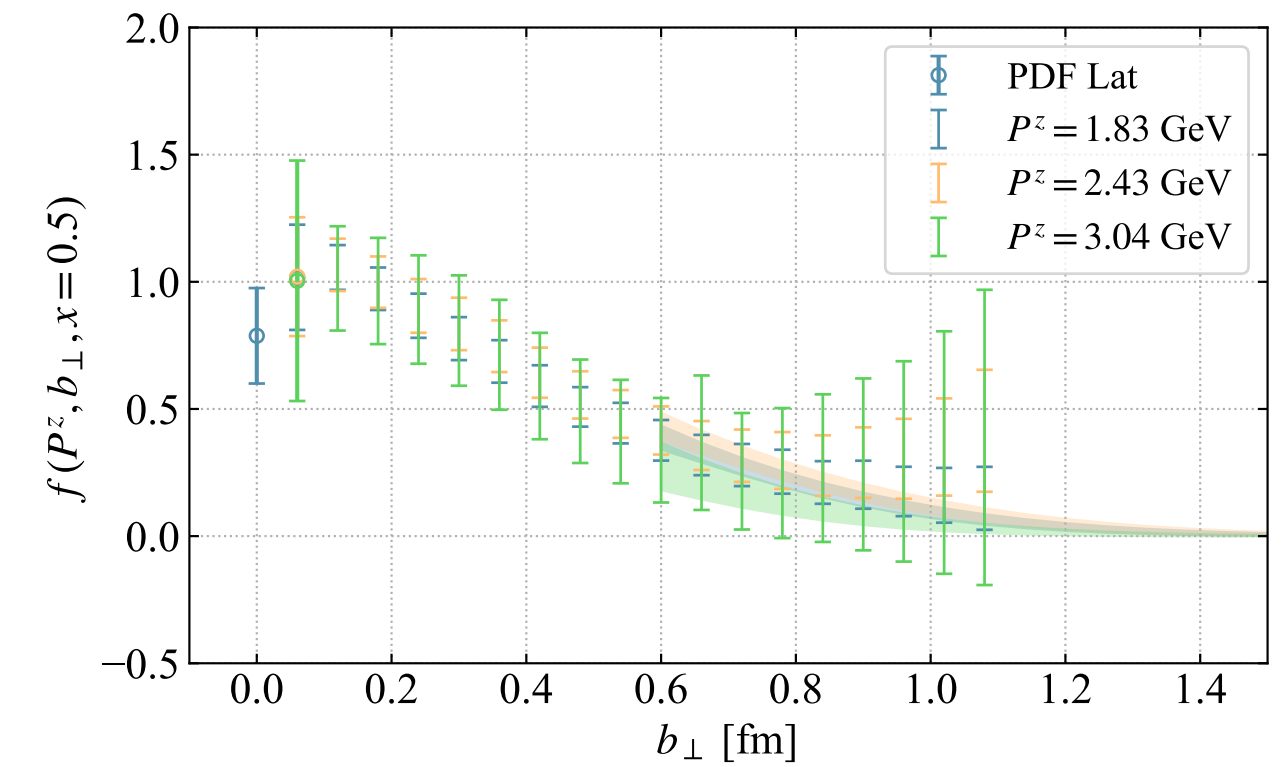
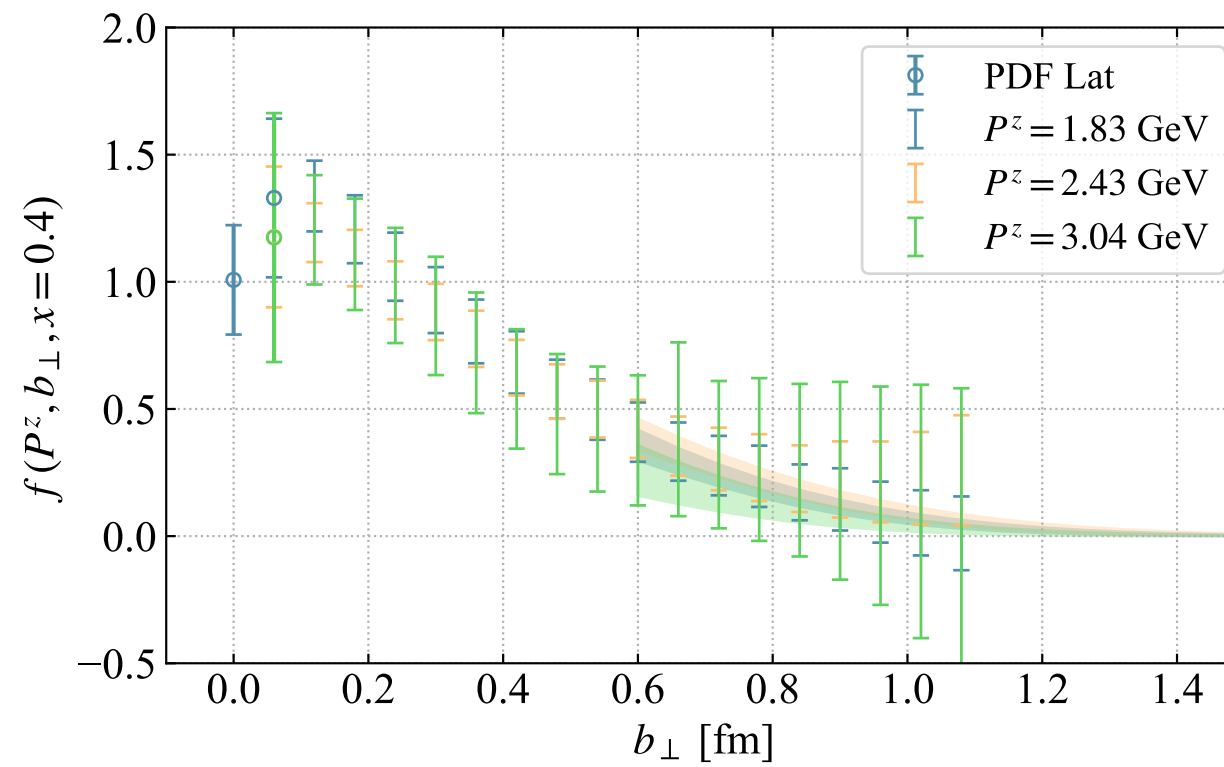
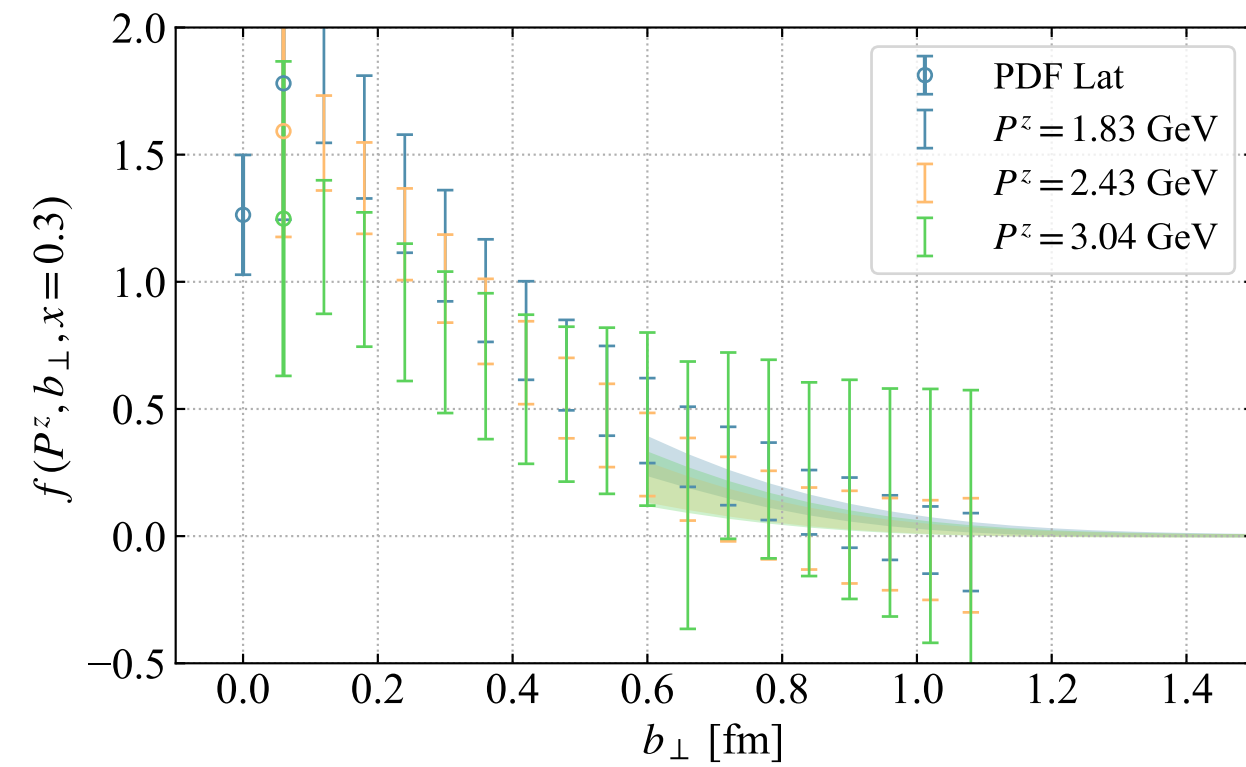
- Thanks to the absence of linear divergence, we can calculate pion TMDPDF up to $b_\perp > 1$ fm;
- When x gets larger, the amplitude of TMDPDF is decreasing, while the the transverse correlation length stays roughly the same;
- When x gets closer to $x = 0.5$, we can find that the variance across different momenta becomes smaller, indicating the suppression of power correction;
- While when x gets closer to $x = 0.5$, the deviation from global analysis becomes larger, which may cause by the fact that the experimental data gives better constraints to the small x region;

Pion TMDPDF in the $k_{\perp}$ Space

- We can give the $k_{\perp}$ -dependence thanks to the good SNR in CG;

- Extrapolate the large $b_{\perp}$ using a simple Gaussian form: $f(b_{\perp}) = Ae^{-mb_{\perp}^2}$;

- Fourier transform to the $k_{\perp}$ space: $\tilde{f}(k_{\perp}) = \int \frac{d^2\vec{b}_{\perp}}{(2\pi)^2} e^{i\vec{b}_{\perp} \cdot \vec{k}_{\perp}} f(b_{\perp}) = \int \frac{d|b_{\perp}|}{2\pi} |b_{\perp}| \cdot J_0(|b_{\perp}| \cdot |k_{\perp}|) \cdot f(b_{\perp})$.



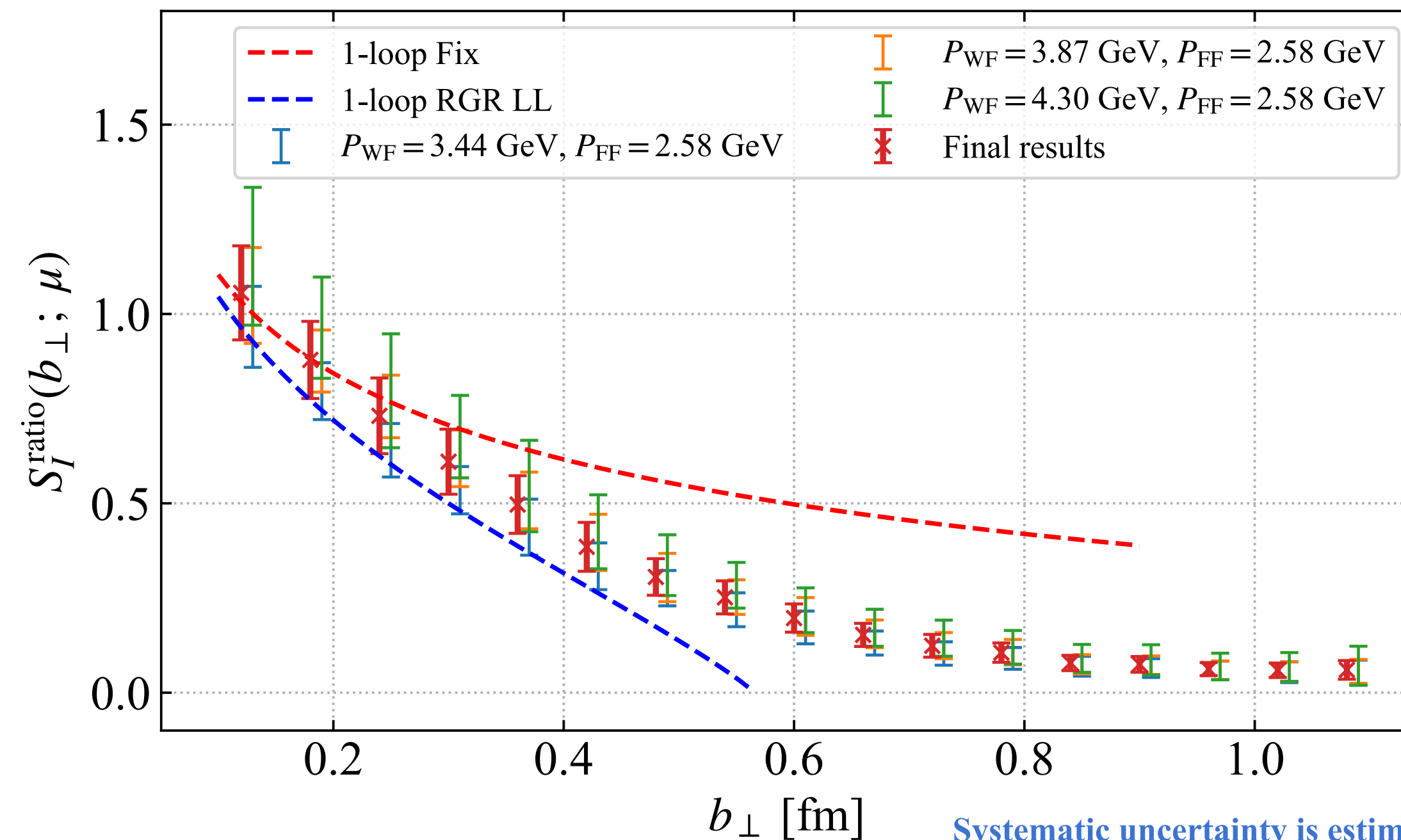
Summary

- This is the **first lattice calculation** of the pion unpolarized TMDPDF within LaMET framework;
- The **novel CG** method is employed to remove the linear divergence, so that to have a good SNR up to $b_{\perp} > 1$ fm;
- The soft function is extracted at **NLL factorization** using RG resummation, the results show consistency with perturbation theory;
- The TMDs, including CS kernel, intrinsic function, TMDWF and TMDPDF are calculated using the same lattice ensemble, and the results show consistency with existing studies, including phenomenology and lattice calculations;
- The outcome of this study highlights the efficacy of the CG quasi-TMD approach in probing the transverse momentum structure of hadrons;
- In the future work, we will apply the CG quasi-TMD approach on nucleon, and the lattice systematics like discretization effects and non-physical pion mass will be investigated in detail.

Backup

Intrinsic Soft Function

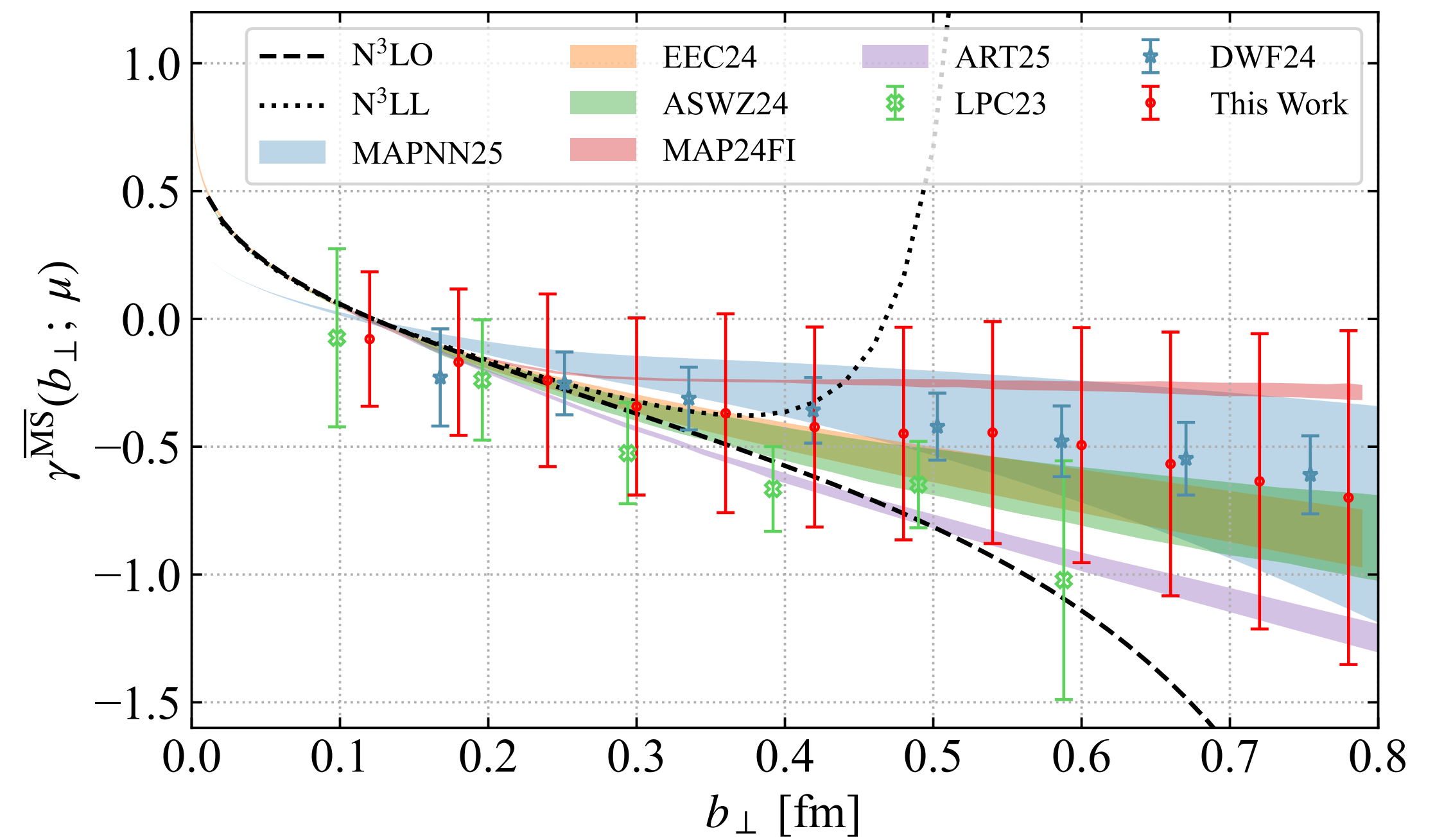
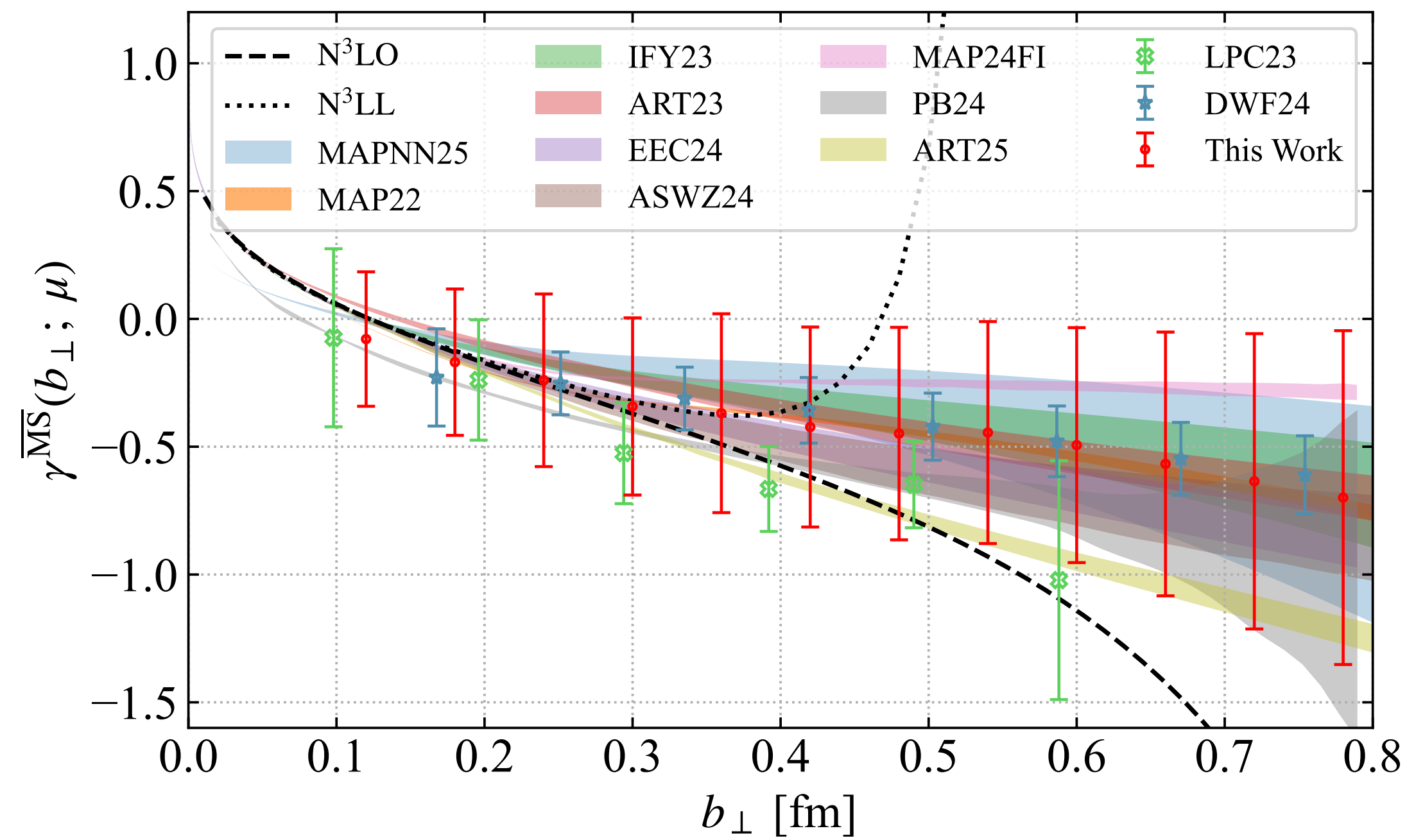
$$S_I(b_\perp; \mu) = \frac{F(b_\perp, P^z)}{\int dx_1 dx_2 H_F(x_1, x_2, P^z; \mu) \tilde{\Phi}^\dagger(x_1) \tilde{\Phi}(x_2)} \text{ with } \tilde{\Phi}(x) \equiv \frac{\tilde{\phi}_\Gamma(x, b_\perp, P^z; \mu)}{H_\phi(x, \bar{x}, P^z; \mu)}$$



Systematic uncertainty is estimated by the spread of the mean values between these three momentum pairs.

- Our lattice results are consistent with the perturbation theory in the small $b_\perp$ regime;
- Different momenta of quasi-TMDWF give consistent results;
- Thanks to the absence of linear divergence, our final results of the intrinsic soft function can go beyond $b_\perp \sim 1 \text{ fm}$.

CS Kernel



Gauge Fixing in Lattice QCD

Continuous Theory

$$F_{\text{CG}}[A, \Omega] \equiv \frac{1}{2} \sum_{\mu=1}^3 \int d^4x A_{\Omega\mu}^a(x) A_{\Omega}^{\mu a}(x)$$

$$\begin{aligned} \delta F_{\text{CG}}[A, \Omega] &= - \sum_{\mu=1}^3 \int d^4x (D_{\mu ab}^{\Omega} \theta_b) A_{\Omega}^{\mu a} \\ &= - \sum_{\mu=1}^3 \int d^4x (\partial_{\mu} \theta_a - g f^{cab} A_{\Omega\mu}^c \theta_b) A_{\Omega}^{\mu a} \\ &= \sum_{\mu=1}^3 \int d^4x \theta_a (\partial_{\mu} A_{\Omega}^{\mu a}) \end{aligned}$$

$$* A_{\Omega\mu}(x) \equiv \Omega^{\dagger}(x) A_{\mu}(x) \Omega(x) + \frac{i}{g} \Omega^{\dagger}(x) \partial_{\mu} \Omega(x)$$

Lattice Theory

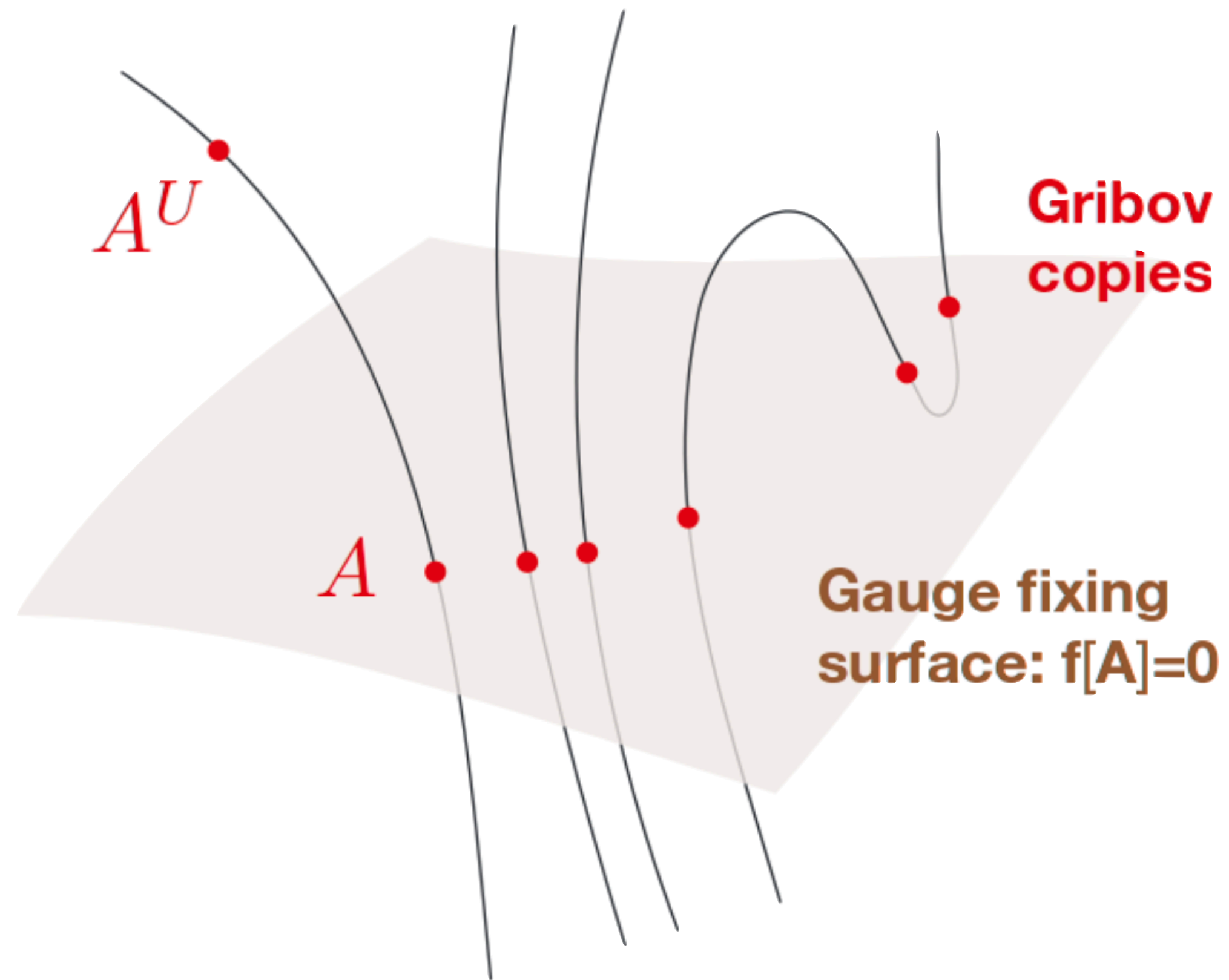
$$F_{\text{CG}}[U, \Omega] \equiv -\Re \left[\text{Tr} \sum_x \sum_{\mu=1}^3 \Omega^{\dagger}(x + \hat{\mu}) U_{\mu}(x) \Omega(x) \right]$$

Find stationary points of the functional value.

Gauge fixing criterion in this work: variation of functional satisfies $\delta F/F < 10^{-8}$.

Gribov Copies

The gauge fixing condition may have many solutions in Lattice QCD.



Ph. D. Thesis of Diego Fiorentini

Criteria of Gauge Fixing

- Variation of the functional

$$\delta F/F < 10^{-8}$$

- Residual gradient of the functional

$$\theta^G \equiv \frac{1}{V} \sum_x \theta^G(x) \equiv \frac{1}{V} \sum_x \text{Tr} \left[\Delta^G(x) (\Delta^G)^\dagger(x) \right], \Delta^G(x) \equiv \sum_\mu \left(A_\mu^G(x) - A_\mu^G(x - \hat{\mu}) \right)$$

